DOI: 10.1111/fima.12335

ORIGINAL ARTICLE

Financial Management WILEY

The effects of political competition on the funding of public-sector pension plans

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Funding information
Office of Tax Policy Research, the Doctoral Studies Office at the University of Michigan, and Villanova University

Abstract

In politically competitive jurisdictions, there can be strong electoral incentives to underfund public pensions in order to keep current taxes low. I examine this hypothesis using panel data for over 2,000 local pension plans from Pennsylvania spanning the period 1985–2017. The results suggest that more politically competitive municipalities tend to have pension plans that are less funded. The effects of political competition are driven by municipalities that have a higher proportion of uninformed voters and are absent for pension plans offered by municipal authorities. The negative relationship between political competition and funding status is present for state pensions as well.

The sky really is falling. The people of Jacksonville need to know this. When they look at the roads that are crumbling. When they look at the neglected infrastructure. When they look at certain ZIP codes that have had little to no investment, promises that were made in years gone by that haven' t been kept. It' s because we don' t have the cash to do it because our budget is being eaten alive by these unfunded pension liabilities. – Mayor of Jacksonville (FL), Lenny Curry[1]

1 | INTRODUCTION

The bankruptcy of cities such as Stockton, San Bernardino, and Detroit has brought the issue of public-sector pension plans into focus. Unfunded pension and health care obligations have been key drivers behind the decisions of these municipalities to file for bankruptcy. However, the size and funding of public-sector pension plans are issues affecting states and municipalities throughout the country.

[1] Source: http://news.wjct.org/post/curry-proposes-extending-sales-tax-pay-pension-debt

Under defined benefit (DB) pension plans, the dominant type of pension in the public sector, state and local governments promise pension benefits that are typically a specific fraction of an employee' s last drawn salary. Sponsoring employers are expected to put money into a retirement trust fund so that, when combined with employees' contributions, the fund will grow sufficiently to provide the promised retirement benefits when the employee retires. In practice, however, large gaps often exist between the assets available in the retirement trust fund and the promises made to employees. Some estimates put the magnitude of unfunded liabilities—the excess of promises over assets—at over $4.5 trillion as of June 2019.[2] The aggregate level of unfunded pension liabilities, however, conceals considerable heterogeneity across plans, the reasons for which have not been adequately explored in the literature.

This paper offers a novel explanation for the existence of the underfunding that focuses on the role of incentives faced by politicians who decide on the terms and conditions of employment for public-sector workers. It proposes that political competition, defined as the lack of a systematic electoral advantage by either political party, plays a key role in the underfunding of pension plans. To the extent that public-sector workers are better informed than workers in the private sector, competition for votes creates incentives for politicians from both parties to offer generous retirement benefits to workers in the public sector and simultaneously, to not fund them fully, in order to avoid raising taxes on workers in the private sector. Thus, in a competitive political environment where there are few partisan voters who support a given party regardless of the economic platforms of these parties, politicians attempt to get every vote possible. As a result, they make promises of generous pension benefits to public-sector workers while lacking the political will to raise taxes high enough to cover the full costs of these benefits. As political competition intensifies, these incentives become sharper, resulting in the prediction that a higher degree of political competition is associated with a decline in the funding status of pension plans *and* an increase in the generosity of such benefits. For purposes of tractability, this paper examines the effects of political competition on one of those dimensions—the funding of pension plans—with the effects of political competition on the generosity of such plans the focus of a complementary paper (Bagchi, 2019).

To test the effects of political competition on funding, I examine local pension plans in the state of Pennsylvania. Pennsylvania provides a rich setting to investigate these issues because its local governments offer over 1,400 retirement systems, which account for more than 40% of all public-employee systems nationally.[3] The existence of such a large number of plans is driven partly by the plethora of local governments and by the fact that unlike other states with a similarly large number of local governments, there has never been a state-wide consolidation of local plans. Given the political landscape of the state, we also observe wide variation in the level of political competition, our independent variable of interest. Finally, I am able to construct a panel data set spanning the period from 1985 through 2017 using reports of the Public Employee Retirement Commission (PERC). This administrative high-quality data set covers the universe of local pension plans and does not suffer from nonresponse bias making it particularly appealing for our use.

The primary contribution of this paper is an empirical analysis of the effects of political competition on the funding of local DB plans in the state, whether offered by municipal governments or municipal authorities. In contrast to the governing body of a municipality that is directly impacted by the outcome of an election and is likely to be responsive to voters, the appointed board of a municipal authority is relatively removed from the electoral will of voters. The insulation of municipal authorities from political influence along with their ability to establish pension plans for their employees subject to the same reporting and funding standards as municipalities enables me to use authorities as a "control group" or placebo. Unlike the negative relationship between political competition and pension plan funding levels I observe for municipal plans, I find no relationship between the level of political competition and funding levels for plans run by municipal authorities. These results are robust to controlling for municipality fixed effects, suggesting that unobserved time-invariant heterogeneity across municipalities (or municipal authorities) are not driving the

[2] Author calculations based on Rauh, J. (June 11, 2019). "Hidden Debt, Hidden Deficits: The 2019 Update," https://www.hoover.org/news/hidden-debt-hidden-deficits-2019-update.

[3] Source: https://www.census.gov/prod/2013pubs/g11-aspp-sl.pdf (reference table 5 a). Of the 3,418 public-employee retirement systems in the United States, 1,422 (or, 41.6%) are local retirement systems from Pennsylvania.

results. In terms of magnitude, a one standard deviation increase in the level of political competition is associated with a long-run decline in the actuarial funded ratio of municipal plans by about 3.4% with *no* effect on plans run by municipal authorities.[4]

I next examine a key implication of the model that the underfunding of pension plans by politicians can persist in equilibrium only if private-sector workers are not fully informed of the benefits that have been promised to public-sector workers. In a test, I split the sample of municipalities into two groups based on the level of voter engagement with the prediction that the effects of political competition are larger in municipalities that have a higher proportion of uninformed voters. Using election turnout of the population as a proxy for voter awareness and engagement, I find that the effects of political competition are indeed larger in municipalities where election turnout is lower than or equal to the median. I find similar results when instead I use residential mobility or educational attainment as proxies of voter awareness. The results from the sample-split analysis suggest that when faced with an electorate that is more aware and informed of the true cost of unfunded pension obligations, politicians are less likely to underfund public pensions.

Finally, to assess the external validity of these results, I estimate a similar set of regressions for state plans using a panel data set spanning the period from 1989 through 2017. The negative relationship between political competition and the funding level of pension plans re-emerges when we use data on state pensions, with the estimated coefficients of political competition similar in magnitude to those obtained for municipal pension plans from Pennsylvania. Those results offer some confidence that the relationship between political competition and fiscal health of public pensions is not unique to Pennsylvania but is likely to be valid in other contexts as well.

This paper relates to a literature in political economy that has examined the effects of political competition on the choice of economic policy. Acemoglu and Robinson (2006) propose that more political competition may intensify political instability and diminish the incentives of elites to implement growth-enhancing reforms. Likewise, Lizzeri and Persico (2005) argue that competition, as measured by the number of parties, may reduce welfare as proliferation in parties may lead to an increase in targeted transfers rather than an increase in public goods.

Unlike these models where political competition is detrimental, Besley et al. (2010) (henceforth BPS) propose a model in which political competition induces parties to adopt policies that promote growth rather than engage in rent seeking. One key difference between BPS (2010) and this paper is that while the former focuses on the choice of policies that are easily visible to voters, this paper focuses on the effect of political competition on the choice of a policy—the funding level of a public pension—that has historically not been visible to the average voter.

This paper also relates to the literature on fiscal illusion—the notion that taxpayers' failure to perceive the full extent of tax burdens can lead to a failure of recognizing the true cost of public services and hence generate significant distortions in fiscal policy (Banzhaf & Oates, 2013). Related to the notion of a fiscal illusion are models of political budget cycles that predict monetary and/or fiscal expansions prior to elections. While the first generation of models (e.g., Nordhaus, 1975) rested on some form of irrationality on the part of voters, subsequent models consider voters as fully rational but imperfectly informed (e.g., Rogoff, 1990; Rogoff & Sibert, 1988). Although empirical tests of the political budget cycle have had mixed results, a common finding has been that higher the degree of transparency, the smaller is the cycle (e.g., Alt & Lassen, 2006; Brender & Drazen, 2005; Shi & , 2006)—a finding that finds support in our context as well.

Lastly, this paper contributes to the literature that has examined the existence of underfunding of public pensions. Theoretical explanations proposed focus on the differences in borrowing costs of citizens from the pension fund relative to the costs of borrowing in the private market (Mumy, 1978) or the desire of current residents to move out of a given jurisdiction and pass on the costs of these pensions to future residents (Inman, 1982). These explanations appear inadequate to explain the variation in the extent of funding found across retirement plans in practice.

[4] A one standard deviation increase in the level of political competition, using the measure defined in Besley, Persson, and Sturm (2010), results if the Democratic vote share goes down from 57.3% (leaning Democratic) to 50% (most competitive), or goes up from 42.7% (leaning Republican) to 50%.

Closest to the idea presented in this paper is Epple and Schipper (1981), who conjecture that increased political competition may induce politicians to underfund pension liabilities, so that they can reduce taxes in the short run;[5] this behavior is rewarded by those voters who are unaware of deferred pension obligations. Although the authors raise this possibility, they conclude otherwise based on their preliminary evidence. They examine variation in house prices within various municipalities in the Pittsburgh Standard Metropolitan Statistical Area (SMSA) and find some evidence that pension obligations are capitalized in house prices and accordingly conclude that these obligations are visible to local residents. Even if we accept their finding that unfunded pension obligations are reflected in house prices, that does not necessarily lead to the conclusion that *voters* are cognizant of unfunded pension obligations.

Also relevant to us in providing a theoretical underpinning for this study is Glaeser and Ponzetto (2014) who posit that pension obligations are shrouded because of lower availability of information about pensions than wages and because of the greater difficulty taxpayers face in understanding the operation of DB plans, in contrast to current compensation.[6] Given voters' lack of understanding of DB pensions, the political process induces an inefficiently high level of provision of retirement benefits. Furthermore, prefunding generous retirement benefits fully would negate the appeal of offering these benefits in the first place as that would require raising taxes to a level that would attract scrutiny and consequently, lower levels of funding become likely as the generosity of benefits increases.

The final paper that this paper relates to and builds on is Elder and Wagner (2015). The authors incorporate electoral uncertainty and political polarization in a model where governments must decide how to allocate tax revenues among two different public goods and funding a pension plan. Parties differ in their preferences over these public goods and the authors prove that as electoral uncertainty or political polarization rise, the funding level chosen for the pension plan declines. Using a panel of 91 state plans for the period from 2001 to 2010, the authors report findings consistent with their model: Higher legislative turnover rates, more electoral competition, and the presence of term limits all lead to lower pension plan funding. A notable difference of this paper from Elder and Wagner (2015) is that while electoral uncertainty is exogenous to their model (an incumbent government has an exogenous probability of losing office), the model sketched in this paper endogenizes it based on the promises made by the two parties that compete for votes.

A number of empirical papers (e.g., Coggburn & Kearney, 2010) also examine possible causes for underfunding but those studies tend to have important limitations. They are often based on cross-sectional analyses at the state level, raising the possibility that unobserved heterogeneity across states might be driving the findings reported in these papers. Additionally, they introduce variables that are likely to be endogenous (e.g., the decision of whether or not to make the actuarially required contribution in Munnell, Aubry, & Quinby, 2010, which examines variation in the funding status of 126 state and local DB pension plans as of 2009). This paper addresses such limitations, in part by constructing a panel data set spanning a longer period of time than in any prior work and introducing municipal fixed effects that account for unobserved heterogeneity across jurisdictions. It is also the first paper to focus exclusively on local pensions and one that critically examines—and finds evidence consistent with—the assumptions that underpin the theoretical model that I lay out in the following section.

The paper proceeds in seven sections. Section 2 develops the hypotheses and provides the outline of a theoretical model, leaving all formal elements and proofs of propositions to the Appendix. Section 3 describes the data sources and the empirical specifications. Section 4 presents our results of the impact of political competition on the funding status of pension plans. Section 5 presents results from sample-split analyses examining the heterogeneity of the relationship between political competition and funding status based on differences in voter awareness. Section 6 discusses the validity of the empirical approach and presents results for the effects of political competition on the funding of state pension plans. Section 7 concludes.

[5] In a somewhat different context, Cohn, Gurun, and Moussawi (2020) note that CEOs with short-term incentives distort project choices.

[6] In support of their claim, they note that salaries of state employees are publicly disclosed annually whereas no such database exists for the accruing pensions of retirees or active members.

2 | HYPOTHESES DEVELOPMENT

In this section of the paper, I develop the hypotheses that focus on the role of political competition in explaining variation in the funding status of public pensions. I do so by outlining a two-period model in which more intense political competition exacerbates the incentives of politicians to not fund the retirement benefits that have been promised to public-sector workers so as to avoid raising taxes. The key assumptions made in constructing the model are that (i) politicians care about voter welfare in addition to winning elections (e.g., Ansolabehere, 2008; Wittman 1977, 1983) and (ii) as conjectured by Epple and Schipper (1981), private-sector voters have imperfect information about the funding and generosity of public-sector pensions (Glaeser & Ponzetto, 2014).[7]

The model assumes the existence of two parties, which differ on a dimension unrelated to their stance on economic issues, which I label as "ideology."[8] Ideology is not amenable to change at will during an election campaign and is assumed invariant over time. Voter heterogeneity is along two dimensions, the first being the level of partisanship and the second being the sector in which they work. I assume that a fraction of the voters are partisans who prefer their respective parties because of ideological considerations and by definition, their utility gain from having their preferred party in office overrides any economic concerns. Without loss of generality, I let the Democrats have the edge among committed voters. I characterize the remaining voters as "swing" and distinguish between them based on their sector of work. A fraction of these swing voters work in the public sector, while the remaining work in the private sector. I take the fraction of partisan voters, the proportion of partisan voters supporting each party, and the fraction of swing voters who work in each sector as exogenous parameters of the model. Political competition is modeled more formally in the Appendix but as can be intuited from this discussion, a higher level of political competition arises when we have fewer partisan voters and when neither party has a surplus of committed partisan voters. In such a case, electoral promises arise endogenously to appeal to swing voters.

In this two-period model, workers in both sectors of the economy work in the first period and retire in the second. While public-sector workers have access to a stream of retirement benefits from the government, private-sector workers' consumption in the second period of the model is driven by their savings during the first period of the model. The timing of the game is as follows: In the first stage of the model, both parties announce platforms simultaneously under uncertainty about an aggregate popularity shock.[9] Next, the aggregate popularity shock is realized as voters consider the platforms announced by the two parties and cast their votes.[10] Finally, the party winning the election assumes office and implements its announced platform and voters make private economic choices in light of the policy chosen.[11]

I assume that the government is constrained to running a balanced budget as far as current compensation of public-sector workers is concerned. It inherits a pension fund at the start of period 1 that is balanced, that is, has enough assets to cover all liabilities and operates under the constraint that the pension fund be balanced at the end of period 2. To keep the model tractable, I focus on a single policy decision by the government and a single economic decision by public- and private-sector workers. The only decision the government makes is the extent, if any, to which it runs a

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[7] Assumption (i) is not critical. Politicians may want to minimize the extent of unfunded liabilities motivated by considerations of self-interest rather than altruism. Politicians generally own residences in the jurisdictions that they govern and may therefore desire to reduce the level of unfunded liabilities simply to avoid any possible capitalization of these liabilities in house prices. Politicians might also care about pension funding because of concerns about employment opportunities after their stint in elected office as they may have fewer opportunities if their tenure is associated with poorly run pension plans. Finally, it may be the case that underfunding a pension plan beyond a certain level attracts the scrutiny of state regulators or holders of municipal debt. Any of those possible considerations would justify why politicians may care about funding a pension plan, and not make decisions based solely on maximizing their electoral prospects. Assumption (ii) is however critical and absent it, private sector workers would be able to perfectly foresee and offset potential future tax hikes by adjusting their level of savings (a Ricardian equivalence–type result).

[8] The political model draws on the model of electoral competition laid out in Persson and Tabellini (2002).

[9] Platforms are assumed to be binding.

[10] I assume full turnout in the model and do not consider abstentions.

[11] Although the model has two periods, I model only one election, that held prior to the first period. Nothing hinges on this. All I require is that the government in office in period 2 honor the pension obligations incurred in period 1.

deficit in the pension fund in the first period of the model, taking as exogenous the level of wages in both sectors and the level of retirement benefits offered in the public sector. The only decision made by workers is how much to save in the period when they are working, taking into account their wages, retirement benefits (if any), consumption preferences, and projected path of taxes.

The model is driven by the imperfect information problem private-sector workers face as they decide how much to save. These workers prefer underfunding because they view it as a reduction in the level of present and future taxes rather than an intertemporal shift of taxes from the present to the future. However, given that benefits promised to public-sector workers are paid out in all states of the world, the consequence of underfunding the pension plan is an increase in taxes in the second period of the model. Unable to anticipate this increase in taxes, private-sector workers end up saving too little in the first period. This decision leads to a suboptimal intertemporal allocation of consumption and hence lower welfare than in a world of full funding or a world in which private-sector workers have perfect information about the funding level of the pension plan.

Parties, which care about voters as well as winning, decide on their electoral platform that involves setting taxes for the first period of the model based on weighing the ego rents they receive from coming to office *and* the decrease in utility for private-sector workers as a result of the government running a deficit in the pension fund. An equilibrium of the model can be represented by a pair of funding levels, which forms a Nash equilibrium in the pre-election game between the two parties, given the equilibrium behavior of voters.

The predictions of the model are that in an environment with minimal political competition, the party with more partisan voters wins even when they fund the pension plan fully and because doing so increases the welfare of private-sector workers, the party chooses to pursue a policy of full funding. As political competition increases, the party with more partisan voters chooses to fund the plan less than fully although it does not still choose the lowest allowable funding level. By contrast, the party that has fewer partisan voters (the underdog) chooses the lowest allowable funding level in order to maximize its chances of winning. As political competition increases, the underdog party increases its chances of winning (and accordingly the probability that the lowest pension plan funding level is chosen goes up) whereas the overdog party reduces its chosen level of funding. As a result, the funding level chosen for the pension plan goes down in equilibrium as political competition increases. At some point however, when political competition gets sufficiently intense, the effect of a further increase in political competition on the funding level is ambiguous.

3 | EMPIRICAL ANALYSIS

3.1 | Institutional context

In order to empirically examine the hypothesis that a higher level of political competition is associated with a lower actuarial funded ratio, I examine local pension plans from the state of Pennsylvania. Pennsylvania provides a rich setting for the empirical analysis as it accounts for two-fifths of the nation' s distinct public sector retirement systems and offers more than thrice the number of retirement systems as that of any other state. The existence of such a large number of retirement systems in Pennsylvania can be attributed to its complex system of local government. General purpose local governments—cities (56), boroughs (958), and townships (1,547)—total approximately 2,600 units. While there are exceptions, cities are governed by a mayor and four council members and boroughs by a council of either three or five or seven members who may be elected either at large or by wards. Townships of the first class are governed by either five commissioners elected at large or up to 15 elected by wards, whereas townships of the second class—the most numerous class of municipalities in the state—are governed by three supervisors who are elected at large.

Most relevant to the discussion at hand, unlike other states with a large number of municipal governments, most general purpose local governments in Pennsylvania establish separate pension plans for their police and nonuniformed employees and are not part of a larger state-wide plan.[12] The advantages of using municipal data to test the hypotheses are the large number of comparable cases that share the same national- and state-level political context (e.g., state tax rates) at the same time they exhibit wide variation on the variables of interest, namely, political competition and pension funding status. In addition, the availability of rich municipal-level data from the Decennial Censuses and the American Community Surveys (ACS) enables me to control for many potentially important municipal characteristics that could influence the funding of public pensions.

Besides municipalities, there are over 1,500 active municipal authorities in the state (DCED, 2015). Municipal authorities typically perform a very limited number of functions such as the provision of parking (e.g., Allentown Parking Authority) or water (e.g., Harrison Township Water Authority). Municipalities justify the provision of services through an authority on multiple grounds. Beyond the single-minded focus that the board of an authority can have on its operations and the fact that many services are provided efficiently only if a large service area spanning multiple municipalities is covered, the governance of municipal authorities may also be more conducive to their operation. Authority board members are appointed by the governing body of the municipalities where they provide service for five-year overlapping terms. Therefore, in contrast to the governing body of a municipality that can change following an election, the appointed board of an authority is relatively insulated from the electoral will of voters.

The insulation of municipal authorities from political influence along with their ability to establish pension plans for their employees subject to the identical reporting and funding standards as municipalities enables me to use authorities as a "control group" and contrast the effects of political competition on municipal plans from the effects on plans run by municipal authorities.[13]

3.2 | Construction of variables and data sources

Data regarding pension plans offered by local governments within Pennsylvania are available from 1985 through 2017 in the form of biennial status reports prepared by the PERC.[14] Status reports include the name of the municipal entity offering the plan, the employee group covered, the actuarial liabilities, actuarial assets, and number of active members in the plan.[15] Using these reports, I construct two variables that are used subsequently: the actuarial funded ratio, defined as the ratio of actuarial assets to actuarial liabilities multiplied by 100, and unfunded liabilities, defined as (actuarial liabilities − actuarial assets).[16]

Constructing measures of political competition at the local level is challenging as there is no central repository for data on municipal elections at either the federal or the state level. The data reside at the county level and few counties have historical election data publicly available. de Benedictis-Kessner and Warshaw (2016) note, for example, that

[12] Larger municipalities may also have a separate pension plan for firefighters. Teachers belong to a separate state-wide system, the Pennsylvania Public School Employees Retirement System (PSERS), and are not a part of this analysis.

[13] Operationalizing this test requires assigning each authority to a single municipality. Doing so is straightforward for an authority that serves a single municipality where the service area of an authority and a municipal boundary perfectly match (e.g., Allentown Parking Authority). In the case of municipal authorities that service more than one municipality (e.g., Valley Joint Sewer Authority), the following rules are used in order to assign an authority to a single municipality. First, I examine the composition of the authority's board and assign the authority to that municipality which has the largest number of seats. If that information is unavailable or inconclusive (e.g., equal number of board seats from two municipalities), I examine how the authority is financed; I assign the authority to whichever municipality (or municipal residents) pays the largest share of expenses associated with the authority's operation. If such information is not available either or inconclusive, I then assign the authority to that municipality which is most populous among all municipalities served by the authority. About half of all municipal authorities match unambiguously with a single municipality. Full details of the assignment are available from the author on request. Table A.1 in the Supplemental Information examines the effects of political competition on the actuarial funded ratio and unfunded liabilities and finds that the null results for authorities are robust to whether the match with a municipality is unambiguous or not.

[14] The PERC was abolished by Act 100 of 2016, and the former commission' s duties were transferred to the Department of the Auditor General. The 2015 and 2017 reports were accordingly sourced from the AG's office.

[15] From 1989 onwards, the reports also provide the unfunded liabilities expressed as a share of payroll.

[16] To take an example, consider a plan with 100 active members whose actuarial liabilities and assets are valued at \$4 million and \$3 million, respectively. For such a plan, the actuarial funded ratio is 3/4, or 75%, unfunded liabilities are (\$4 − \$3) million or \$1 million, and unfunded liabilities per active member are \$1 million/100 = \$10,000.

"it was nearly impossible to find election results for most cities with fewer than 75,000 people..." Only seven municipalities in Pennsylvania exceed 75,000 people, with the median-sized municipality having fewer than 2,000 residents. In light of these significant challenges to obtaining local election data, I construct measures of political competitiveness at the local level using vote shares for the two parties for all state and national offices for which elections are held on a state-wide basis, namely, U.S. President, U.S. Senator, Governor,[17] Attorney General, Auditor General, and Treasurer.[18] Underlying data on votes cast for these offices for candidates from various parties are available at the municipal level from 1980 through 2016 in successive issues of the Pennsylvania Manual. Because the results for a particular candidate in any one election cycle may have a large idiosyncratic component to it, I average the Democratic vote share[19] across all elections held within a given time period to any of the six offices in constructing the average Democratic vote share for that time period. In Section 6.1, I offer evidence that the approach of using elections to national- and state-level offices to construct measures of competitiveness at the local level is a reasonable one.

The key measure of political competition I use in the paper is that used in BPS (2010); political competition for municipality *m* over a time period, indexed *t* is defined as: $PC_{mt} = -|D_{mt} - 0.5|$, where D_{mt} is the average Democratic vote share in municipality *m* over period *t*. In the empirical analysis, I start off with a parsimonious specification that includes the measure of political competition, municipality and period fixed effects, and employee-group dummies. In subsequent specifications, I include numerous demographic controls at the municipal level that might affect pension plan funding level such as per capita income. These variables are drawn from the 1980, 1990, 2000 Decennial Censuses, and the 2007–2011, 2011–2015, and 2013–2017 ACS.[20] Summary statistics for all variables are presented in Table 1.

The table indicates the large amount of variation in the dependent variables: the actuarial funded ratio and unfunded liabilities, even though these variables have been winsorized at the 5% and 95% levels.[21] One way in which I address this wide variation is that in one of our robustness checks, I take the log of unfunded liabilities making it less sensitive to outliers. I also note the large variation in the level of political competition from −0.427 (corresponding to a Democratic vote share of 0.073 in Bryn Athyn Borough, Montgomery County, for the years between 1980 and 1984—least competitive) to −0.000 (corresponding to a Democratic vote share of 0.500).[22]

The summary statistics also raise the issue of whether pension plans are overfunded on average given that the mean exceeds 100% for municipal plans and plans run by municipal authorities and the median funded ratio is about 100% for plans run by both types of entities. The actuarial funded ratio is based on numbers as reported by the municipalities themselves and discount future actuarial liabilities at an interest rate ranging between 5.5% and 8.0%, the median being 7.0%. Assuming an interest rate of 5% for discounting actuarial liabilities, reflecting yields on high-grade corporate bonds (as is required for private-sector DB plans), result in the median actuarial funded ratio declining from 98.97% to 77.33%. If an interest rate of 3.5% were used instead reflecting the nominal yield on long-term Treasury bonds and the essentially guaranteed nature of these pension promises, the median actuarial funded ratio would decline further to 64.14%, reflecting considerable underfunding in the median municipal plan.[23]

[17] There is no separate general election for the Lt. Gov.'s post as the Gov. and Lt. Gov. run on one ticket at that stage.

[18] As BPS (2010) note, name recognition of candidates for down-ballot offices is typically very low among voters, making it likely that measures of political competition based on races for these offices is driven largely by party attachment of voters rather than the popularity of individual politicians.

[19] Defined as votes cast for Democrats/(votes cast for Democrats + votes cast for Republicans).

[20] Linear interpolation using the Decennial Censuses is used prior to 2000. The 2000 Census is the source of data for 2000; the 2007–2011 ACS being centered around 2009 is the source of data for that year and likewise the 2011–2015 and 2013–2017 ACS are the source of data for 2013 and 2015. Years between 2001 and 2008 are interpolated using the 2000 Census and the 2007–2011 ACS, years between 2010 and 2012 are interpolated using the 2007–2011 and the 2011–2015 ACSs, and the year 2014 is interpolated using the 2011–2015 and the 2013–2017 ACSs.

[21] Results are similar if I winsorize either at the 1% to 99% level or the 2.5% to 97.5% or the 10% to 90% level and are available on request.

[22] These most competitive municipalities are Castle Shannon Boro., Allegheny County for the years between 1980 and 1984; Indiana Boro, Indiana County; Pulaski Twp., Lawrence County; and Nazareth Boro., Northampton County for the years between 1996 and 2006.

[23] These back-of-the-envelope calculations assume that the weighted average duration of pension liabilities is 13 years, in the middle of the range suggested by the literature (Novy-Marx & Rauh, 2011; Rauh, 2017; Waring, 2004a, 2004b).

TABLE 1 Summary statistics

Variable	Units	Mean	Median	Standard deviation	Minimum	Maximum
Pension plan characteristics						
For municipal plans:						
Actuarial funded ratio	In percent terms	112.94	98.97	50.63	54.06	240.26
Unfunded liabilities	In dollars	(3,615)	1,343	38,400	(82,035)	61,697
per active member						
Interest rate	In percent terms	6.88	7.00	0.820	5.50	8.00
For plans run by municipal authorities:						
Actuarial funded ratio	In percent terms	102.91	97.18	35.39	54.06	240.26
Unfunded liabilities	In dollars	1,505	1,731	22,657	(82,035)	61,697
per active member						
Interest rate	In percent terms	6.48	6.00	0.769	5.50	8.00
Controls at the municipal level						
Households that are owner-occupied	In percent terms	72.91	73.85	11.75	55.33	90.98
Population aged 65 or older	In percent terms	15.66	15.70	4.34	7.82	22.88
Unemployment rate	In percent terms	6.06	5.45	2.75	2.49	13.14
Per capita income	In dollars	24,326	23,196	5,429	15,603	35,092
Ratio of debt to taxes	In absolute terms	1.18	0.323	1.79	0.00	6.24
Political variables						
Average Democratic vote share	As a fraction	0.472	0.454	0.127	0.074	0.869
Political competition	As defined in BPS (2010)	−0.108	−0.097	0.073	−0.427	−0.000

Summary statistics for actuarial funded ratio and unfunded liabilities per active member are based on biennial data from 1985–2017 from the PERC. Summary statistics for interest rate are based on biennial data from 2003–2013, also from the PERC. Percentage of households that are owner-occupied, percentage of the population aged 65 or older, unemployment rate, and per capita income are from the 1980, 1990, and 2000 Decennial Censuses and the 2007–2011, 2011–2015, and 2013–2017 ACS. Ratio of debt to taxes is based on annual data available from 1986–2018 from the DCED. All these variables have been winsorized at the 5% and 95% levels. Lastly, average Democratic vote share and measures of political competition are based on elections to all national and state-level offices held between 1980 and 2016 and are constructed using successive issues of the Pennsylvania Manual.

3.3 | Empirical specification

Panel data on the funded ratio and the size of unfunded liabilities are available on a biennial basis over the period from 1985 to 2017. The primary empirical approach used is ordinary least squares in which I include all observations corresponding to DB plans run by municipalities and estimate the effects of political competition on the funding of these municipal DB plans. I then estimate a *separate* set of regressions for all DB plans run by municipal authorities and offer those results as a contrast to the effects of political competition for municipal plans. The empirical specification is:

$$F_{imt} = \alpha + \beta_1 * PC_{m(t-l)} + \beta_2 * C_{it} + \beta_3 * Z_{m(t-l)} + \lambda_m + \gamma_t + \varepsilon_{imt}, \quad (1)$$

where:

- F_{imt} is the dependent variable; for example, the actuarial funded ratio for plan *i* in municipality *m* (or a municipal authority serving municipality *m*) averaged over a period indexed by *t*;
- PC_{mt} is a measure of political competition in the municipality *m* for period *t*, with *l* the lag length;
- C_{it} are a set of dummy variables indicating which group of employees are covered by the plan (e.g., policemen or nonuniformed personnel);
- Z_{mt} are time-variant controls at the municipal level listed below, also introduced with a lag;
- λ_m are municipal fixed effects and γ_t are period fixed effects; and
- ε_{imt} is the error term.

What is of primary interest is the coefficient on political competition, $\widehat{\beta_1}$. In the empirical analysis, I start off with a parsimonious specification that only includes a measure of political competition, in addition to municipality and period fixed effects, and employee-group dummies. A richer specification includes time-varying controls at the municipal level that might affect the funding of pension plans—the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the local unemployment rate, and the log of per capita income. The first control is included because owners may have a longer time horizon than renters, who are more transient, and we may therefore expect homeowners to exercise more discipline on politicians who make decisions about public-sector benefits (Fischel, 2001). Thus, municipal pension plans are expected to be more well-funded on average in jurisdictions where a larger fraction of households are owners. I include the percentage of population aged 65 or older as a control for the age structure of the population as municipalities with a larger fraction of older voters may be more willing to simply pass on these obligations to future generations. The local unemployment rate proxies for local economic conditions as municipalities experiencing high levels of fiscal stress may find it hard to fund their pension plans. I also control for the log of per capita income as more prosperous municipalities may find it easier to raise the tax revenues necessary to fund pension promises that have been paid to public-sector workers. In the most complete specification, I also include the size of municipal debt as a fraction of the taxes a municipality collects. This control offers a snapshot of municipal fiscal health and is constructed using financial reports prepared on an annual basis by the Department of Community and Economic Development (DCED) of Pennsylvania. This choice of control variables is influenced by the prior literature (Coggburn & Kearney, 2010; Eaton & Nofsinger, 2008; Munnell et al., 2010) and data availability. I cluster standard errors at the municipality level throughout to account for potential correlated shocks that generates intertemporal correlation in the error terms.

A critical issue for the empirical specification is the question of how many time periods to aggregate the data to (and therefore implicitly what lag to introduce). However, it is not one that admits an easy answer. The pension data are available from the PERC reports on a biennial basis and that is the highest frequency over which the regressions laid out in Equation (1) could be estimated. However, given that a plan's actuarial liabilities and assets only change slowly over time, those variables and their ratio—the actuarial funded ratio—display considerable persistence. Accordingly it may be a while before increases in political competition at the municipal level translate to drops in pension plan funding and estimating regressions based on the biennial frequency of the data may fail to capture this relationship. On the other hand, estimating regressions at a frequency other than that generated by the data may seem a bit ad hoc as one can split this period spanning 33 years in several different ways. Deciding on how to split the sample period from 1985 to 2017 involves balancing these multiple considerations.

In the end, I decide to split the sample period into three periods of 11 years duration each with period 1 corresponding to years 1985–1995, period 2 corresponding to years 1996–2006, and period 3 corresponding to years 2007–2017, with the independent variable of interest—political competition—and all control variables lagged by one period. Thus, the actuarial funded ratio averaged over the years 1996–2006 is regressed on the average level of political competition constructed on the basis of state and national elections held between 1985 and 1995 (along with other control variables computed for the identical period) while the actuarial funded ratio averaged over the years

2007–2017 is regressed on the average level of political competition computed on the basis of elections held between 1996 and 2006.[24]

This decision to split the data into three time periods of 11 years duration can be justified based on (i) the slow-changing nature of the dependent variable noted previously, (ii) the institutional context of the state, and (iii) a review of the literature. The single-most common form of municipal government in Pennsylvania is a township of the second class, with 1,454 such townships among the roughly 2,600 municipalities. Townships of the second class do not have a mayor and are instead governed by a board of three supervisors with terms staggered such that the voters of each township elect one supervisor to serve for a term of six years.[25] Also as noted earlier, municipal authorities are governed by an appointed board where members have overlapping five-year terms.[26] The fact that terms are six years long for the most common class of municipality or five years long for municipal authorities, combined with the fact that these terms are staggered, suggests splitting the data into periods longer than the four-year cycles, which tend to dominate the U.S. political landscape.

A review of the literature also suggests longer lags of the order of 10 years. For example, in examining public-sector debt dynamics, Antonini, Lee and Pires (2013) note: "The analysis shows debt dynamics in the EU 10 are complicated, involving important intercountry interactions and *protracted adjustment periods of about 10 years* (emphasis added)." The unfunded pension liabilities that we analyze in this paper are akin to debt; they can only change incrementally over time. Similarly, Pickering and Rockey (2011) in analyzing the determinants of the size of government in OECD countries note: "Because in reality there are substantial lags between preferences, as expressed in the ideology data, and policy enacted by government, in the regression analysis below, we use a moving average of the previous ten years of the Manifesto Research Group data.… Taking the average of the past ten years of data helps to lessen concerns about endogeneity; the ideology measures now substantially predate the observations on government size."[27] In our context too concerns about endogeneity carry some weight; had we been using contemporaneous (or near contemporaneous) measures of political competition, there would be the concern that poor pension plan funding levels in some jurisdictions induce increases in political competition and accordingly, causality runs from low pension funding to higher political competition rather than in the manner proposed in the paper. Accordingly constructing averages over longer intervals of time and then introducing the variables of interest with a lag[28] reduces such concerns.[29]

The decision to split the sample period into three periods of 11 years duration each is consequential; by doing so we implicitly assume that there is a considerable lag before a change in political competitiveness of a municipality manifests itself in the form of a change in the fiscal health of a municipal pension plan. However, given the nature of the dependent variable and the staggered terms of municipal boards, this decision is sensible. Also importantly, our results are not particularly sensitive to this assumption and as the results provided in Table 3 indicate, our key finding that higher levels of political competition are associated with lower actuarial funded ratios for municipal plans holds up regardless of how we split the sample period up.

[24] As the election data start in 1980, I regress the funded ratio for a pension plan averaged over the years 1985–1995 on the level of political competition constructed on the basis of state and national elections held in 1980, 1982, and 1984.

[25] As per 53 P.S. 65403 Second Class Township Code. The code also specifies the rules by which voter approval can be obtained for expanding the board of supervisors from three to five members.

[26] This description regarding the structure of the board for municipal authorities is typical: "The governing body of the [Middletown Township Sewer] Authority is a Board consisting of five members appointed by the Township Council. The terms of the members of the Board are staggered so that the term of one member expires each year. Members of the Board may be reappointed" (https://middletowndelcopa.gov/sewer).

[27] In a follow-up paper, Pickering and Rockey (2013) in explaining variation in the size of U.S. state government posit "a link between lagged ideology measures and current policy" because as they note "In practice, bureaucratic inertia might mean that it takes time before the full impact of ideological shifts manifest themselves in policy changes. At any point in time government size is likely to be cumulatively a function of ideology in several past years rather than exactly mirroring the current zeitgeist."

[28] Introducing a lag is also in line with the literature on political budget cycles where the use of retrospective voting models is common (e.g., Fiorina, 1981; Kramer, 1971).

[29] While driven by data availability, using vote shares from state-wide and national elections, rather than municipal elections, may also lessen endogeneity concerns; voters are less likely to blame state (or national) officials, rather than local officials, for poorly performing pension plans. Motivated by similar concerns, I introduce a robustness check (RC1) in Table A.4, which only uses data from Presidential elections to construct measures of political competition.

4 | EXAMINING THE FUNDING STATUS OF PENSION PLANS

In this section, I present estimates of the effects of political competition on the funded ratio of DB pensions operated by various municipalities and municipal authorities within Pennsylvania. I start off with results from a baseline specification, which are followed by a host of robustness checks.

4.1 | Results from the baseline specification on actuarial funded ratio

The data available from the biennial reports of the Pennsylvania PERC from 1985 through 2017 enable us to analyze the effects of political competition on the funding status of local pension plans in the state. In Table 2, I present the regression results obtained by estimating specification (1) with actuarial funded ratio as the dependent variable, with columns (1) through (3) providing the coefficient on political competition for municipalities and columns (4) through (6) providing the coefficient for municipal authorities. Columns (1) and (4) correspond to the most parsimonious specification and only includes political competition, municipality and period fixed effects, and dummy variables for the various employee groups covered by the pension plans. Columns (2) and (5) introduce the time-variant controls from the Census that control for homeownership, the age structure of the population, the local unemployment rate, and per capita income. Columns (3) and (6) augment those by including a municipal-level fiscal control, namely, the ratio of debt to taxes collected by the municipality.[30] All tables that include results for municipalities and municipal authorities follow a similar pattern.

The negative and statistically significant coefficients on political competition in columns (1) through (3) suggest that a higher level of political competition is associated with a decline in the funded ratio for municipal plans. To provide a sense of magnitude of these effects, if the level of political competition were to increase by one standard deviation,[31] the funded ratio for the average municipal pension plan would decline by about 3.4%.[32] In contrast, based on examining the coefficients in columns (4) through (6), we do not see any statistically significant effects of political competition on the funding status of plans run by municipal authorities.[33]

4.2 | Robustness checks

In this section, I examine the robustness of the finding that political competition is associated with a decline in the funding status for municipal plans but not for plans run by municipal authorities. These checks are geared toward establishing that our results are robust (1) to operationalizing the dependent variable differently, (2) to alternative ways of cutting the data, and (3) to distinguishing between the effects of political competition and straight-up partisanship.

4.2.1 | Alternative measures of the funding status of DB pension plans

Two municipalities of similar size could have pension plans with similar funded ratios and yet the magnitude of their unfunded liabilities could differ, if pension plans operated by one municipality are more generous than pension plans

[30] Results are similar if we instead use a measure of the size of government, such as revenues or expenses per capita.

[31] A one standard deviation increase in the level of political competition, using the measure defined in BPS (2010), would result if the Democratic vote share were to go down from 57.3% (leaning Democratic) to 50% (most competitive), or conversely, go up from 42.7% (leaning Republican) to 50% (most competitive).

[32] For example, the coefficient on political competition is −47.28 in column (1), whereas a one standard deviation change in political competition equals 0.0728. Thus, a one standard deviation increase in political competition corresponds to a decline in the actuarial funded ratio by $-47.28 \times 0.0728 = -3.44\%$.

[33] The standard errors are however large enough that I cannot rule out the possibility that the coefficients on political competition for municipal plans are equal to the corresponding coefficients for municipal authority plans, with the p-values that examine if the coefficients are different or not equal to 0.1511, 0.1290, and 0.1198, respectively.

TABLE 2 Effect of political competition on actuarial funded ratio using the absolute difference of the democratic vote share from 50% as the measure of political competition

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Political competition	−47.28***	−47.07***	−47.07***	−4.067	−1.496	−0.546
	(12.09)	(11.98)	(11.99)	(36.61)	(36.69)	(36.57)
Percent housing units		0.590*	0.589		−0.579	−0.595
owner-occupied		(0.358)	(0.359)		(1.026)	(1.020)
Percentage of population		−0.446	−0.446		−0.374	−0.354
aged 65 or older		(0.376)	(0.376)		(0.943)	(0.951)
Unemployment rate		0.0202	0.0199		−1.550	−1.537
		(0.515)	(0.515)		(1.469)	(1.478)
Log of per capita income		−23.19*	−23.17*		−25.14	−26.60
		(13.05)	(13.06)		(33.05)	(32.73)
Ratio of debt to taxes			0.0629			0.807
			(0.474)			(1.369)
Municipality and period FE	✓	✓	✓	✓	✓	✓
Employee-group dummies	✓	✓	✓	✓	✓	✓
Number of observations	5,600	5,600	5,600	847	847	847
Number of pension plans	2,052	2,052	2,052	352	352	352
Number of municipalities	1,272	1,272	1,272	341	341	341
(or municipal authorities)						
R^2	0.58	0.58	0.58	0.66	0.67	0.67

Robust standard errors clustered at the municipality level are in parentheses;
*$p < 0.10$,
**$p < 0.05$,
***$p < 0.01$.
Regressions estimated on all public-sector defined benefit pension plans from Pennsylvania for the period 1985–2017, which is split into three time periods (1985–1995, 1996–2006, and 2007–2017). The dependent variable, the actuarial funded ratio, is defined as the percent of pension liabilities funded. The measure of political competition used is that defined by BPS (2010), namely, $PC_{mt} = -|0.5 - D_{mt}|$. Controls for class of employee covered, municipality and period fixed effects are included in all specifications. Municipal demographic controls included in columns (2), (3), (5), and (6) are the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the local unemployment rate, and the log of per capita income. Municipal fiscal control included in columns (3) and (6) is the ratio of debt to taxes collected by the municipality. The dependent variable and all control variables have been winsorized at the 5% and 95% levels.

operated by the other municipality. Therefore, an alternative approach of quantifying the fiscal health of DB pensions is to examine the level of unfunded liabilities. Normalizing unfunded liabilities by some measure of size is important as larger municipalities are likely to have larger unfunded liabilities, all else equal. Hence, I use a number of variables to normalize unfunded liabilities dividing it variously by (i) the number of active members in the plan, (ii) the size of the plan's payroll, and (iii) population. Those results are presented in Panels A through C of Table A.2 with the coefficient on political competition for municipalities in columns (1) through (3) and that for municipal authorities in columns (4) through (6). Given the wide dispersion in the level of unfunded liabilities across municipalities, I also use the log of unfunded liabilities per active member and the log of unfunded liabilities per capita as the dependent variable in Panels

D and E, respectively. In the interest of space, only the coefficients on political competition are presented in this (and subsequent) tables with complete results available on request.

As suggested by the coefficients in columns (1) through (3) of Table A.2, there is a positive and statistically significant relationship between political competition and the size of unfunded liabilities for municipal pensions, regardless of how those liabilities are normalized. In terms of the magnitude of these effects, a one standard deviation increase in the level of political competition is associated with an increase in unfunded liabilities by about $3,500 when expressed on a per member basis, about 7% as a share of the payroll, and about $5 in terms of unfunded liabilities per capita.[34] In contrast, there is no evidence that political competition increases the size of unfunded liabilities for municipal authorities, regardless of how these liabilities are normalized, with the coefficients in fact negative in all five panels.

4.2.2 | Alternative ways of cutting the data

The checks in this subsection are geared toward establishing the robustness of the negative effects of political competition on the funded ratio of municipal plans to alternative ways of cutting the sample period. While the reasons for cutting the data spanning 33 years into three periods of 11 years duration each and then introducing all variables with a lag were noted in Section 3.3, alternative ways of splitting the sample period are certainly possible. For example, one could simply estimate the regressions at the lowest frequency possible—biennially—based on the frequency at which the pension data are available from the PERC. Alternatively, because terms of office are four years long in townships of the first class, boroughs, and cities but six years long in townships of the second class, one could argue for aggregating the data over periods that are either four years or six years long. One might also agree that while longer time periods are necessary, aggregating the data into four eight-year periods or two 16-year periods—the longest span possible is preferred. Table 3 addresses this issue by estimating specification (1) by splitting the sample into periods that are 2 years long (Panel A), 4 years long (Panel B), 6 years long (Panel C), 8 years long (Panel D), and 16 years long (Panel E), and introducing all variables with a lag.

As the results in Table 3 suggest, our finding that there is a negative relationship between municipal-level political competition and the funding of pension plans run by that jurisdiction holds up regardless of how our sample period is cut and the data averaged over those periods. However, as can be seen from the table, the coefficients on political competition increase in magnitude as we move from Panel A to Panel E, with the largest effects of political competition arising when the sample period is split into two periods of 16 years duration each.[35] Moreover, the coefficients on political competition for municipal authorities are statistically insignificant all throughout, with the coefficient positive in all panels but Panel E.

4.2.3 | Examining whether partisanship matters for funding

The narrative presented thus far in the paper suggests that there is a causal link between political competition and the actuarial funded ratio, with higher levels of political competition leading to declines in pension plan funding. One may however wonder if there is a direct effect of partisanship per se, and whether those effects can be distinguished from the effects of political competition itself. I take two approaches to address these questions. Given that in the regressions presented, I have not included a control for the average Democratic vote share, the first approach is to simply introduce that variable in the regressions as it enables us to separately identify the effect of an increase in

[34] For example, in column (1) of Panel A, the coefficient on political competition is 48.12 and a one standard deviation change in political competition equals 0.07246. As notes to Table A.2 indicate, unfunded liabilities per member have been rescaled by dividing by $1,000 for ease of presentation. Thus, the estimated effect of political competition on unfunded liabilities per member = $48.12 \times 1000 \times 0.07246 =$ $3,487.

[35] Note that the *R*-squared also tends to go up as we move from Panel A to Panel E.

TABLE 3 Effect of political competition on actuarial funded ratio using alternative ways of cutting the data

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Panel A: Splitting the sample into periods that are 2 years long						
Political competition	−10.42*	−9.516*	−9.435*	6.814	13.78	13.77
	(5.349)	(5.346)	(5.343)	(14.94)	(14.40)	(14.43)
Number of observations	28,459	28,459	28,459	3,992	3,992	3,992
R^2	0.53	0.53	0.53	0.54	0.55	0.55
Panel B: Splitting the sample into periods that are 4 years long						
Political competition	−18.45***	−17.27***	−17.21***	14.94	18.85	18.51
	(6.610)	(6.598)	(6.591)	(18.26)	(18.21)	(18.10)
Number of observations	12,589	12,589	12,589	1,784	1,784	1,784
R^2	0.56	0.56	0.56	0.59	0.60	0.60
Panel C: Splitting the sample into periods that are 6 years long						
Political competition	−19.86***	−18.61**	−18.56**	5.415	10.20	10.15
	(7.690)	(7.637)	(7.619)	(22.76)	(22.68)	(22.45)
Number of observations	10,857	10,857	10,857	1,547	1,547	1,547
R^2	0.54	0.54	0.54	0.58	0.58	0.58
Panel D: Splitting the sample into periods that are 8 years long						
Political competition	−26.56***	−25.19***	−25.02***	7.224	12.01	12.05
	(9.033)	(9.029)	(9.018)	(27.81)	(26.96)	(26.76)
Number of observations	7,407	7,407	7,407	1,089	1,089	1,089
R^2	0.57	0.57	0.57	0.60	0.61	0.61
Panel E: Splitting the sample into periods that are 16 years long						
Political competition	−64.37**	−62.94**	−63.10**	−35.66	−23.53	−20.06
	(29.26)	(29.12)	(29.16)	(89.34)	(90.42)	(93.56)
Number of observations	3,849	3,849	3,849	601	601	601
R^2	0.63	0.63	0.63	0.72	0.72	0.73
Municipality and period FE	✓	✓	✓	✓	✓	✓
Employee-group dummies	✓	✓	✓	✓	✓	✓
Municipal demographic controls		✓	✓		✓	✓
Municipal fiscal control			✓			✓

Robust standard errors clustered at the municipality level are in parentheses;
*$p < 0.10$,
**$p < 0.05$,
***$p < 0.01$.

Regressions estimated on all public-sector defined benefit pension plans from Pennsylvania for the period 1985–2017, which is split into periods of varying lag lengths as noted for each panel. The measure of political competition used is that defined by BPS (2010), namely, $PC_{mt} = -|0.5 - D_{mt}|$. Controls for class of employee covered, municipal and period fixed effects are included in all specifications. Municipal demographic controls included in columns (2), (3), (5), and (6) are the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the local unemployment rate, and the log of per capita income. Municipal fiscal control included in columns (3) and (6) is the ratio of debt to taxes collected by the municipality. The dependent variables and all control variables have been winsorized at the 5% and 95% levels.

Democratic support from an increase in the level of political competition. While Panel A of Table 4 provides the baseline specifications for ease of comparison, Panel B now additionally controls for the average Democratic vote share.

The results in Panels A and B suggest that an increase in political competition is associated with a decline in the actuarial funded ratio for municipal plans, *regardless of whether we control for partisanship or not*, with the coefficients in Panel A indistinguishable from the corresponding coefficients in Panel B. Moreover, the coefficient on the average Democratic vote share is statistically insignificant all throughout suggesting that partisanship does not have an *independent* influence on the funded ratio. These results can be captured in the form of Figure 1, which highlights that (1) the negative relationship between political competition and the funded ratio holds up regardless of whether we control for the Democratic vote share or not and (2) there is no direct relationship between partisanship and the funded ratio, once political competition is controlled for.

A second attempt at addressing this question of the direct effect of partisanship involves operationalizing political competition differently and introducing a linear and squared term for the average Democratic vote share in the same specification. If political competition makes pension plans less funded, then one would expect to see a negative coefficient on the linear term and a positive coefficient on the squared term generating an U-shape. This is, in fact, what I find with this alternative operationalization of political competition in Panel C. Based on my estimates, the Democratic vote share, which corresponds to the lowest level of funding varies in a range from 42.2% to 61.3%, with the 95% confidence intervals including 50% in all specifications. Taking the point estimates literally, pension plan funding level decreases as the Democratic vote share increases to about 51.4% and then they increase.[36] Thus, Panel C also confirms the negative and statistically significant relationship between political competition and the actuarial funded ratio of municipal pension plans. The results in this table suggest that partisanship does not directly impact the funded ratio and the effects that we observe and document in this paper stem from variations in the level of political competition, rather than from variations in partisanship.

Overall the results presented in this section in Tables 2 through 4 offer robust evidence that an increase in the level of political competition is associated with a decline in the funded ratio of municipal plans with no effect on the funding status of plans run by municipal authorities. The results are also robust to numerous additional checks provided in Appendix Table A.4, such as constructing measures of political competition based only on Presidential elections, including municipality-specific time trends, and controlling for the number of active members in the plan.

5 | EXAMINING THE ASSUMPTIONS MADE IN THE CONSTRUCTION OF THE THEORETICAL MODEL AND ITS IMPLICATIONS

5.1 | A sample-split test examining differential effects based on voter awareness

A critical assumption of the theoretical model is that private-sector workers are not fully informed of the retirement benefits that have been promised to public-sector workers. This is in line with the assumptions in Glaeser and Ponzetto (2014) who also argue that pension obligations are shrouded because of lower availability of information about pensions than wages and because of the greater difficulty of understanding the accrual of pension obligations in contrast to current compensation. This suggests an empirical test. By splitting the sample of municipalities into two groups based on the level of voter awareness and information, we can examine if the effects of political competition are larger in municipalities that have a higher proportion of uninformed voters. To operationalize these tests, I consider variables that likely reflect variations among the residents of a municipality in their ability to understand the complexities of local public finance and their incentives to do so.

[36] For example, using the coefficients in column (1), the level of Democratic vote share at which funding is minimized = 184.1/(2 × 186.6) = 49.3% and the p-value that the implied point of maximum equals 50% is 0.8530.

TABLE 4 Effect of political competition on actuarial funded ratio for municipal plans: examining the role of partisanship

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			**Coefficient for plans run by municipal authorities**		
Panel A: Measure of Political Competition: Absolute Difference of Democratic Vote Share from 50%						
Political competition	−47.28***	−47.07***	−47.07***	−4.067	−1.496	−0.546
	(12.09)	(11.98)	(11.99)	(36.61)	(36.69)	(36.57)
Number of observations	5,600	5,600	5,600	847	847	847
R^2	0.58	0.58	0.58	0.66	0.67	0.67
Panel B: Measure of political competition: Absolute Difference of Democratic Vote Share from 50%, augmented with Democratic vote share						
Political competition	−48.05***	−44.23***	−44.26***	−1.822	2.145	2.660
	(12.79)	(12.75)	(12.77)	(35.03)	(35.30)	(35.17)
Coefficient on Democratic	2.318	−8.587	−8.500	−48.51	−40.89	−38.12
vote share	(13.94)	(14.89)	(14.96)	(41.03)	(47.39)	(47.49)
Number of observations	5,600	5,600	5600	847	847	847
R^2	0.58	0.58	0.58	0.67	0.67	0.67
Panel C: Measure of political competition: Linear and Squared term for Democratic vote share						
Coefficient on a linear term for	−184.1***	−181.0***	−181.2***	−65.38	−44.30	−42.16
Democratic vote shared	(44.94)	(44.59)	(44.66)	(153.1)	(154.0)	(154.1)
Coefficient on a squared term	186.6***	172.7***	173.1***	16.88	3.725	4.413
for Democratic vote share	(46.95)	(46.95)	(47.13)	(137.9)	(140.3)	(138.3)
Number of observations	5,600	5,600	5,600	847	847	847
R^2	0.58	0.58	0.58	0.67	0.67	0.67
Municipality and period FE	✓	✓	✓	✓	✓	✓
Employee-group dummies	✓	✓	✓	✓	✓	✓
Municipal demographic controls		✓	✓		✓	✓
Municipal fiscal control			✓			✓

Robust standard errors clustered at the municipality level are in parentheses;
*$p < 0.10$,
**$p < 0.05$,
***$p < 0.01$.
Regressions estimated on all public-sector defined benefit pension plans from Pennsylvania for the period 1985–2017. The dependent variable, the actuarial funded ratio, is defined as the percent of pension liabilities funded. The measure of political competition used in Panels A and B is that defined by BPS (2010), namely, $PC_{mt} = -|0.5 - D_{mt}|$. Controls for class of employee covered, municipality and period fixed effects are included in all specifications. Municipal demographic controls included in columns (2), (3), (5), and (6) are the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the local unemployment rate, and the log of per capita income. Municipal fiscal control included in columns (3) and (6) is the ratio of debt to taxes collected by the municipality. The dependent variables and all control variables have been winsorized at the 5% and 95% levels.

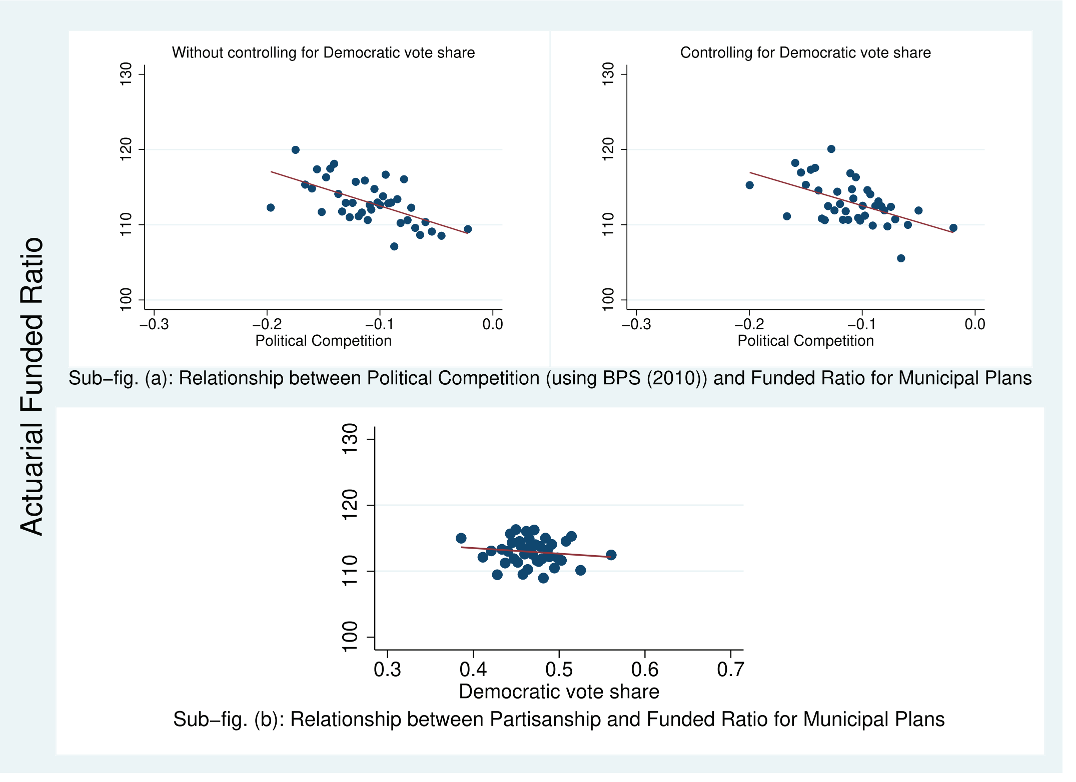

FIGURE 1 How partisanship moderates the relationship between political competition and the actuarial funded ratio These plots were drawn using the Stata command, binscatter, with 40 bins used in all cases: Sub-figure (a) corresponds to the regression results presented in column (3) of Panels A and B of Table 4 in which I regress the actuarial funded ratio on political competition, municipality and period fixed effects, employee-group dummies, and all demographic and fiscal controls. Sub-figure (b) corresponds to regressing the actuarial funded ratio on the Democratic vote share, controlling for political competition, and the identical set of controls as listed above. Data on actuarial funded ratio of municipal defined benefit pension plans are available from biennial reports of the Pennsylvania Public Employee Retirement Commission (PERC). This variable has been winsorized at the 5% and 95% levels. Data on Democratic vote share have been constructed using results of all national- and state-level elections held in Pennsylvania as described in the text. Election results are available from successive issues of the Pennsylvania Manual. Political competition is defined as the absolute difference of the Democratic vote share from 50% following Besley et al. (2010). [Color figure can be viewed at wileyonlinelibrary.com]

The first variable I use for the sample-split analysis is turnout in national and state elections as voting is a tangible measure of citizens' participation in the political process and thought of as "the most basic form of civic engagement" (Smith, Schlozman, Verba, & Brady, 2009). Columns (1)–(3) present the coefficient on political competition for municipalities where turnout in the national and state elections was lower than or equal to the median, while columns (4)–(6) present the coefficient for municipalities where turnout was higher than the median.[37] The last three columns examine if the coefficients in columns (1)–(3) are statistically different from those in columns (4)–(6).

The second variable that I use for the sample-split analysis is the proportion of residents who report staying in the same house five years ago (on the Decennial Census)/one year ago (on the ACS).[38] The response to these questions

[37] To reduce any concerns about the sorting of residents over time, I classify municipalities based on turnout in elections over the period 1980–1984 and *preserve* the sorting of municipalities over time.

[38] The specific question asked on the Decennial Census was: "Did this person live in this house or apartment 5 years ago?" The ACS asks an identical question except that it replaces "5 years" with "1 year."

about residence five (or one) years ago can be used to capture the degree of residential mobility. The relationship of this variable with voter awareness and engagement can be traced to Putnam (1995) who notes that moving, "like frequent repotting of plants, tends to disrupt root systems, and it takes time for an uprooted individual to put down new roots" (p. 30). Therefore, in Panel B of Table 5, I split the sample of municipalities based on the percentage of residents who have stayed in the same house over either a five-year or a one-year horizon. Columns (1) through (3) present the coefficient on political competition for municipalities where the percentage of residents staying in the same house is less than or equal to the median while columns (4) through (6) present the coefficient for municipalities where the percentage of residents staying in the same house is more than the median.

The third and final variable I use for the sample split is education—more specifically, the percentage of the population that has a high school degree (or higher). This variable is motivated by Almond and Verba (1989, 1st ed. 1963), who see education as a crucial determinant of "civic culture" and note that "The uneducated man or the man with limited education is a different political actor from the man who has achieved a higher level of education" (p. 315). Likewise, Delli Carpini (2000) suggests that education is the strongest single predictor of political knowledge.[39] Thus, Panel C reports results from splitting municipalities into two groups based on the percentage of the population that possesses at least a high school degree, using data from the 1980 Census.

Results in Panel A of Table 5 suggest that the effects of political competition are larger in municipalities where turnout in national and state elections is lower than or equal to the median. Similarly, results in Panels B and C indicate that the effects of political competition are larger in municipalities (b) where the percentage of residents staying in the same house is less than or equal to the median and (separately) (c) where the percentage of adult residents with a high school degree (or higher) is less than or equal to the median,[40] with the differences statistically significant in all three panels. While none of these variables perfectly capture voters' degree of comprehension of the nuances of local public finance, they are likely to correlate with voter awareness and engagement.

Importantly, these results are not simply driven by places with lower awareness also having lower pension plan funding levels; municipalities where turnout and educational attainment are less than (or equal to) the median in fact have higher pension plan funding levels *on average* compared to places where turnout and educational attainment are higher than the median. Moreover, given that the results arise in a specification that includes municipality fixed effects and that they hold up to controlling for variations in per capita income and home ownership, the effects we observe are not driven by differences in observable characteristics across municipalities either. The results suggest that when faced with an electorate that is more aware and informed of the true cost of unfunded public pensions, politicians are less likely to underfund pension obligations. These results are consistent with one of the key assumptions of the model as it pertains to limited information on the part of voters and provides an empirical basis for that assumption.[41]

5.2 | Falsification test using municipal debt levels

As per the mechanism proposed in this paper, the opacity of pensions and voters' inability to fully comprehend them drive the relationship between political competition and the funding status of municipal pensions. Enabling politicians to act strategically with electoral considerations in mind is the lack of checks and balances that lets politicians pass on the true costs of pension obligations onto future taxpayers. That fact suggests a falsification exercise: examine the relationship between political competition and the level of debt held by a municipality. Municipal governments in Pennsylvania, like local governments elsewhere, operate under a balanced budget requirement. Moreover, deficits are

[39] Akhmedov and Zhuravskaya (2004) use the regional share of population with higher education in examining how variations in voter awareness dampen the political budget cycle in the regions of Russia between 1996 and 2003. Likewise, Janků and Libich (2019) use the proportion of the labor force with tertiary education in constructing their measure of voter awareness and find that political budget cycles occur in all countries except those with well-informed voters.

[40] Results are similar if instead of using the percentage of adult residents with a high school diploma, I split them based on the percentage of residents with a bachelor's degree (or higher) and are available on request.

[41] Appendix Tables A.5, A.6, and A.7 demonstrate the robustness of the sample split results for alternative lag lengths.

TABLE 5 Sample-split test examining the effects of political competition on funded ratio, based on differences in voter awareness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Sample of municipal plans split using voter turnout in national and state elections									
	Percentage of residents who turnout to vote <= Median			Percentage of residents who turnout to vote > Median			p-value that coefficients differ across samples		
Political competition	−95.24***	−93.04***	−93.11***	15.02	15.20	15.20	<0.0001	<0.0001	<0.0001
	(19.22)	(19.62)	(19.64)	(15.86)	(15.77)	(15.77)			
Number of observations	2,645	2,645	2,645	2,665	2,665	2,665			
R^2	0.59	0.59	0.59	0.56	0.56	0.56			
Panel B: Sample of municipal plans split using residential mobility									
	Percentage of residents who stayed in the same house <= Median			Percentage of residents who stayed in the same house > Median			*p*-value that coefficients differ across samples		
Political competition	−57.63***	−58.15***	−58.18***	−17.10	−12.76	−12.51	0.0728	0.0414	0.0401
	(17.89)	(17.86)	(17.89)	(18.09)	(17.62)	(17.60)			
Number of observations	2,675	2,675	2,675	2,635	2,635	2,635			
R^2	0.52	0.52	0.52	0.61	0.61	0.61			
Panel C: Sample of municipal plans split using educational attainment of the population 25 years and older									
	Percentage of residents with a high school degree <= Median			Percentage of residents with a high school degree > Median			*p*-value that coefficients differ across samples		
Political competition	−76.83***	−69.29***	−69.28***	−6.568	−6.030	−5.939	0.0015	0.0055	0.0055
	(19.36)	(20.06)	(20.03)	(15.82)	(16.14)	(16.16)			
Number of observations	2,624	2,624	2,624	2,686	2,686	2,686			

(Continues)

TABLE 5 (Continued)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
R^2	0.61	0.61	0.61	0.52	0.52	0.52			
Municipality and period FE	✓	✓	✓	✓	✓	✓			
Employee group dummies	✓	✓	✓	✓	✓	✓			
Municipal demogr. controls		✓	✓		✓	✓			
Municipal fiscal control			✓			✓			

Robust standard errors clustered at the municipality level are in parentheses;
$^{*}p < 0.10$,
$^{**}p < 0.05$,
$^{***}p < 0.01$.
The p-value in columns (7) – (9) is generated from statistical tests which examine whether the coefficient on political competition in columns (1) through (3) differs from corresponding coefficients in columns (4) through (6). Regressions are estimated on all municipal defined benefit pension plans from Pennsylvania for the period 1985–2017 and include municipality and period fixed effects. Plans run by municipal authorities are not included in these estimations. The dependent variable, the actuarial funded ratio, is defined as the percent of pension liabilities funded. The measure of political competition used is that defined by BPS (2010), namely, $PC_{mt} = -|0.5 - D_{mt}|$. Data on votes cast in all national and state elections which are used in splitting the sample are sourced from successive issues of the Pennsylvania Manual and the population data come from the 1980 Decennial Census as described in footnote (38) Municipal demographic controls included are the percentage of households that are owner-occupied, percentage of the population aged 65 or older, local unemployment rate, and log of per capita income. Municipal fiscal control included is the ratio of debt to taxes collected by the municipality. The dependent variable and all control variables have been winsorized at the 5% and 95% levels.

likely to be visible to voters and financial market participants, unlike pensions which, until recently, were not a part of the criteria used by credit rating agencies in rating municipal debt.[42] Therefore, a priori we would expect little to no relationship between municipal-level political competition and the level of debt held by the corresponding municipal government.

Municipal debt levels are therefore what I examine in Table A.8. As debt will be higher in absolute terms for larger municipalities, I normalize it in a number of different ways: by (i) population in columns (1) and (2), (ii) municipal revenues in columns (3) and (4), (iii) municipal expenses in columns (5) and (6), and (iv) municipal taxes in columns (7) and (8). I present results obtained using the BPS (2010) measure of political competition in Panel A and augment that by adding a control for the Democratic vote share in Panel B. In Panel C, I introduce a linear and squared term for Democratic vote share to explore the effects of political competition with an alternative operationalization.

As the results in Table A.8 indicate, we find no evidence of a relationship between the intensity of political competition and the level of debt held by municipal governments, regardless of how I normalize debt levels or how I operationalize political competition. This result is reassuring; it suggests that the negative relationship between political competition and funded ratios that has been the focus of the paper is driven by the difficulty that voters have in understanding the long-term commitments associated with DB pensions rather than being driven by other municipal characteristics such as general financial distress or fiscal mismanagement.

6 | DISCUSSION

6.1 | Reasonableness of using national and state-election data for constructing measures of local competitiveness

Because of data limitations, I have used data on national and state elections to construct measures of political competition at the local level. A sense for how reasonable that assumption is can be gauged by investigating the correlations between the limited data available for local races and races to national and state-level offices for the same time period. Local election data, obtained through filing Right-to-Know requests with various County Boards of Elections, are available for a total of 240 municipalities across five counties of Pennsylvania for the 1980s.[43] Table A.11 presents a series of regressions that examine the correlation between the average Democratic vote share in municipal races and that in national and state elections. I examine the strength of this relationship using the average Democratic vote share based on national-level offices (President and U.S. Senate) in Panel A, races for down-ballot offices (Attorney General, Auditor General, and Treasurer) in Panel B, races for Governor in Panel C, and finally, races to all national and state-level offices in Panel D. Columns (1)–(3) consider a pooled ordinary least squares (OLS) approach whereas columns (4)–(6) introduce county fixed effects. Because voters may have personal knowledge of the candidates to local races in smaller municipalities, I also split the sample and present results for municipalities with population lower than or equal to the median in columns (2) and (5) whereas results for municipalities that are larger than the median are presented in columns (3) and (6).

As the results across all panels and columns of Table A.11 illustrate, there is a positive and statistically significant relationship between the average Democratic vote share in municipal elections and the average Democratic vote share

[42] For example, it was not until April 2013 that Moody' s "announced its final approach to the way it will analyze and adjust pension liabilities as part of its credit analysis of state and local governments" reflecting their view that "pension obligations are a significant source of credit pressure for governments and warrant a more conservative view of the potential size of the obligations" (https://www.moodys.com/research/Moodys-announces-new-approach-to-analyzing-state-local-government-pensions--PR_271186).

[43] The five counties are: Bucks, Chester, Dauphin, Lancaster, and Lehigh. These counties were not chosen at random among the 67 counties in the state. Instead several Right-to-Know requests for local election data were filed with the 18 counties that have the most pension plans. The five chosen were among the most responsive in terms of providing the election data, going back to the 1980s, a period during which these data were not recorded or archived electronically.

for elections held to national and state-level offices.[44] These results suggest that the approach of using elections to national and state offices to construct measures of competitiveness at the local level is a reasonable one.

The second approach I use to examine the validity of using elections to national and state offices for constructing measures of local competitiveness is to analyze the representation of Democrats on municipal councils at a point in time and the Democratic vote share for national and state races held around that point. I operationalize this by examining the relationship between the share of council seats held by Democrats in 2012 following the November 2011 elections and the average Democratic vote share for elections to all national and state offices held in 2010 or 2012. More specifically, I look at elections to the office of the President (Panel A) and all down-ballot offices held in 2012 (Panel B), elections to the U.S. Senate that were held either in 2010 or 2012 (Panel C), and elections for Governor held in 2010 (Panel D). As before, I obtain results with and without county fixed effects and by splitting the sample based on the size of the municipal population.

As the results across all four panels of Table A.12 illustrate, there is a positive and statistically significant relationship between the average Democratic vote share for elections held to national and state-level offices and Democratic representation on municipal councils. This relationship appears in specifications with and without county fixed effects, and it shows up for smaller as well as larger municipalities. These patterns, together with the results in Table A.11, suggest that constructing measures of competitiveness based on the average Democratic vote share for national and state offices offers a reasonable picture of the dynamics of local politics in Pennsylvania.

6.2 | Supportive evidence using results for state pension plans

With respect to the validity of these findings beyond Pennsylvania, I note that preliminary work conducted using data on state pension plans support the conclusions of this paper. Using panel data on 85 DB plans from the Wisconsin Legislative Council for 1989 to 2017 and a measure of political competition constructed using the closeness of all state-wide races[45] I find that as the level of political competition in a state goes up, the actuarial funded ratio of plans offered by that state declines. These results are obtained using a specification mirroring those used for municipal pension plans and include state and period fixed effects, in addition to a number of controls, which may influence a state's funding of pension plans such as its unemployment rate, the ratio of active members to retirees, and the level of professionalization of the state legislature (Squire, 2012, 2017). I present results using the negative of the absolute value of the deviation of the Democratic vote share from 50% in Panel A and augment that by controlling for the Democratic vote share in Panel B to address the question of whether there is any direct effect of partisanship. In Panel C, I introduce a linear and squared term for Democratic vote share to explore the effects of political competition with an alternative operationalization.

As we can see from the results in Table 6, the negative relationship between political competition and the funding level of public pensions re-emerges when we use data on state pensions regardless of the precise manner in which we operationalize political competition and regardless of the controls we include, with coefficients on the variables capturing the intensity of political competition similar in magnitude to those reported earlier. As before, the coefficient on Democratic vote share is statistically insignificant in the specifications in Panel B confirming that partisanship does not have a direct impact on the funding status of public pensions. Likewise, the results in Panel C confirm the hypothesis that pension plan funding declines as Democratic vote share increases, reaches its minimum at Democratic vote share of about 53%, and then increases as Democratic vote share increases beyond that point.[46] Tables A.9 and A.10 confirm

[44] Given the few counties, I do not cluster standard errors at the county level and instead report standard errors that are robust to heteroskedasticity. However, the inferences are unchanged if we use the wild bootstrapping method proposed by Cameron, Gelbach, and Miller (2008) and implemented by Judson Caskey using cgmwildboot.

[45] I am grateful to Jim Snyder for sharing this data set, which updates the data in Ansolabehere and Snyder (2002).

[46] For example, using the coefficients in column (1), the level of Democratic vote share at which funding is minimized = $391.2/(2 \times 369.0) = 53.0\%$ and the p-value that the implied point of maximum equals 50% is 0.2492.

TABLE 6 Effect of political competition on actuarial funded ratio for state pension plans

	(1)	(2)	(3)
Panel A: Measure of political competition: Absolute difference of Democratic vote share from 50%			
Political competition	−61.68*	−63.23**	−54.00*
	(31.65)	(31.23)	(29.96)
Number of observations	146	146	146
R^2	0.52	0.52	0.53
Panel B: Measure of political competition: Absolute difference of Democratic vote share from 50%, augmented with Democratic vote share			
Political competition	−78.18**	−77.93**	−67.01**
	(35.55)	(34.90)	(32.80)
Coefficient on Democratic vote share	−28.85	−29.42	−24.73
	(23.21)	(24.87)	(21.66)
Number of observations	146	146	146
R^2	0.53	0.53	0.54
Panel C: Measure of political competition: Linear and squared term for Democratic vote share			
Coefficient on a linear term for	−391.2*	−392.2*	−313.3
Democratic vote share	(213.2)	(212.3)	(198.6)
Coefficient on a squared term	369.0*	368.3*	294.4
for Democratic vote share	(200.8)	(199.3)	(189.5)
Number of observations	146	146	146
R^2	0.51	0.52	0.53
State and period FE	✓	✓	✓
Plan-specific controls		✓	✓
State socioeconomic controls & fiscal control			✓

Robust standard errors clustered at the state level are in parentheses;
*$p < 0.10$,
**$p < 0.05$,
***$p < 0.01$.
Regressions estimated on all state defined benefit pension plans for the period 1989–2017, which is split into three time periods (1989–1998, 1999–2008, and 2009–2017) largely following the splits used for municipal plans. Political competition, along with all control variables, are averages for the previous period; thus they are introduced with a lag. The dependent variable, the actuarial funded ratio, is defined as the percent of pension liabilities funded and calculated for all plans within a state. The measure of political competition used is that defined by BPS (2010), namely, $PC_{mt} = -|0.5 - D_{mt}|$. This measure is based on data shared by Jim Snyder, which are an update of the data used in Ansolabehere and Snyder (2002) in which the authors examine data on all statewide partisan elected offices between 1942–2000. State and period fixed effects are included in all specifications. Plan-specific controls include the coverage of employees under Social Security and the ratio of active members to retirees. State fiscal and socioeconomic controls included are the median deviation from the state's long-term unemployment rate, the coverage of employees under collective bargaining, average state income tax rate, and the professionalization score of the state legislature.
Notes regarding source of data for control variables: Unemployment statistics for each state are drawn from BLS and a long-run average is constructed by averaging the unemployment rate over the longest period for which state-level unemployment data are available (from 1976 onwards). The specific control variable used is the median deviation of the state unemployment rate over a ten-year period from its long-run average. Coverage in the public sector by collective bargaining is drawn from Unionstats.com. The series on collective bargaining by sector of the economy, at the state level, is available only from 1983 onward. State personal income tax rate is drawn from NBER's Taxsim Calculator. I use the maximum personal income tax rate that would be levied on wage income. Even though the average public sector worker is unlikely to be in the highest personal income tax bracket, the tax rate applicable to her is endogenous to her wages and so it is acceptable to use the highest income tax rate as an instrument for the income tax rate that would be applicable to her. The measure of professionalism in the legislative body that can be compared across states and over time are available for 1979, 1986, 1996, 2003, 2009, and 2015 and sourced from Squire (2012, 2017). I use linear interpolation to obtain scores for all years and construct period averages based on that series. Finally, the number of active members and retirees is available for each plan from reports of the Wisconsin Legislative Council.

that this pattern of results holds up when we consider alternative ways of cutting the data or when we split the sample using turnout and residential mobility as proxies of voter awareness.[47] These results offer some confidence that the relationship between political competition and the fiscal health of public-sector pensions is not unique to Pennsylvania but is likely to be valid in other contexts too.

7 | CONCLUSIONS

This paper suggests that political competition plays a key role in influencing the level of funding of DB pension plans offered by municipal governments to their public-sector employees. In their desire to win re-election, politicians in jurisdictions that are politically competitive promise generous benefits to public-sector workers and then fail to make the actuarial contribution necessary to fund them fully in order to avoid having to raise taxes. The results presented in this paper support this hypothesis and demonstrate that an increase in political competition is associated with a decline in the funding level of pension plans. The results are robust to controlling for municipal and period fixed effects, suggesting that unobserved time-invariant heterogeneity across municipalities or aggregate time trends are not driving these results. This central finding is robust to different measures of the funded status of plans, to various ways of cutting the sample period, and to controlling for partisanship.

In contrast to the robust effects of political competition on the funding of municipal DB plans, I fail to find any evidence of a relationship between the level of political competition and the funding level of pension plans offered by municipal authorities. This null result is likely driven by the fact that members of municipal authority boards are appointed and hence far removed from the electoral will of voters, at least when compared to mayors and council members who make decisions on municipal pensions. The different effects of political competition for municipal authorities when compared with municipalities are in line with papers that have examined differences between appointed and elected officials, including Besley and Coate (2003) who find that public elected utility regulators are more pro-consumer in their regulatory policies and Lim (2013), who finds that the harshness of sentences awarded by elected judges is strongly related to the political ideology of the voters in their districts, while that of appointed judges is not. Indeed, the idea that holding bureaucrats responsible for complex policy areas directly accountable to voters or allowing them to be subject to political influence can result in lower levels of performance as suggested by Whalley (2013) for city treasurers and Ornaghi (2019) in the case of municipal police departments appears relevant here. In our context, based on the evidence presented in this paper, introducing a degree of political insulation between voters and officials responsible for making decisions on public pensions may therefore be an idea worth pursuing.

Given the magnitude of unfunded liabilities and their trajectory, reforming their pension plans has become a matter of first-order importance for policy makers in state and local governments. Reforms to reduce pension liabilities have been enacted in the wake of the Great Recession or continue to be on the table in state and local governments across the country of various political proclivities, in recognition of the fact that the costs they face for these benefits exceed what they are willing or able to pay. An understanding of the complex issues around public-sector pensions, toward which this paper has taken a step, can contribute to the development of such reforms and constrain the ability of politicians to kick the can down the road and pass on the costs of current labor services to future taxpayers. Moving from DB plans that are susceptible to political influence to defined contribution plans that are found to be less susceptible to such influence (Bagchi, 2019) may be one step in that direction. Additionally, as reporting standards issued by the Governmental Accounting Standards Board (GASB) change to better reflect the true magnitude of pension promises[48] and as credit rating agencies start taking pension obligations into account more fully in their

[47] I obtain similar results if I use data from the State Integrity Investigation project used in Xu (2017) to split the sample. The effects of political competition are larger in states where citizens find it harder to access information on pension funds “within a reasonable time period and at no cost.” Those additional results are available on request.

[48] https://www.cpajournal.com/2017/05/08/state-local-government-pensions-crossroads/.

determination of municipal debt ratings,[49] the ability of politicians to obscure the true burden of pension obligations may go down in the future.

Beyond its policy implications, the paper contributes to the broader field of political economy by focusing on the effects of political competition on the choice of a policy that has historically not been visible to the average voter. It demonstrates that political competition systematically alters the behavior of politicians when in office and induces them to make decisions that are suboptimal for society in the long run. However, as results from the sample-split and falsification exercise in Section 5 indicate, voter awareness as well as institutional checks and balances such as a balanced budget requirement can moderate some of the negative effects associated with political competition. Although these ideas were developed in the specific context of public-sector pensions, the notion that political competition may promote behavior oriented toward the short-term on issues that are less salient to voters, may be of much broader relevance than simply the context examined here. Researchers should explore the role of political competition in other settings and see if introducing it in their models can help them make better sense of their phenomena of interest.

ACKNOWLEDGMENTS

I thank Sharon Bogden of the Department of General Services of Pennsylvania, Bernard Kozlowski of the Pennsylvania Public Employee Retirement Commission, Rhonda R. Newton of the Pennsylvania Heritage Foundation, and James M. Snyder Jr. for help in obtaining data. I also thank Subhayu Bandyopadhyay, Sourav Bhattacharya, Tilman Borgers, Richard Boylan, Sebastien J. Bradley, Spencer T. Brien, Charles C. Brown, Traviss Cassidy, Julie Berry Cullen, David J. Denis, Raymond Fisman, Kyle Handley, James R. Hines Jr., Jeffrey L. Hoopes, Lawrence W. Kenny, Erasmus Kersting, Christopher Kilby, Manoj Kulchania, Scott Masten, Rabih Moussawi, Dana M. Muir, Michael Pagano, Leslie E. Papke, Giacomo Ponzetto, Stephen Salant, Nathan Seegert, Dan Silverman, Jagadeesh Sivadasan, Joel Slemrod, Mehmet S. Tosun, Ugo Troiano, Rodrigo Wagner, and seminar participants at the University of Michigan, International Institute of Public Finance Annual Congress, and National Tax Association Conference for helpful comments. Data acquisition for the project was made possible in part by grants from the Office of Tax Policy Research, the Doctoral Studies Office at the University of Michigan, and Villanova University. All links were active as of 06/10/2020. All errors remain my own.

REFERENCES

Acemoglu, D., & Robinson, J. A. (2006). Economic backwardness in political perspective. *American Political Science Review*, *100*(1), 115–131.

Akhmedov, A., & Zhuravskaya, E. (2004). Opportunistic political cycles: Test in a young democracy setting. *The Quarterly Journal of Economics*, *119*(4), 1301–1338.

Alt, J. E., & Lassen, D. D. (2006). Fiscal transparency, political parties, and debt in OECD countries. *European Economic Review*, *50*(6), 1403–1439.

Almond, G. A., & Verba, S. (1989). (1st ed. 1963). *The Civic Culture: Political Attitudes and Democracy in Five Nations*. London: Sage Publications.

Ansolabehere, S., & Snyder, J. M. Jr. (2002). The incumbency advantage in U.S. elections: An analysis of state and federal offices, 1942–2000. *Election Law Journal*, *1*(3), 315–338.

Ansolabehere, S. (2008). Voters, candidates and parties. In B. Weingast & D. Wittman (eds.), *Oxford handbook of political economy*, (29–49). New York: Oxford University Press.

Antonini, M., Lee, K., & Pires, J. (2013). Public sector debt dynamics: The persistence and sources of shocks to debt in 10 EU countries. *Journal of Money, Credit and Banking*, *45*(2-3), 277–298.

Bagchi, S. (2016). *The effects of political competition on the funding and generosity of public-sector pension plans*. Working paper. Retrieved from: https://ssrn.com/abstract=2633565.

Bagchi, S. (2019). The effects of political competition on the generosity of public-sector pension plans. *Journal of Economic Behavior & Organization*, *164*, 439–468.

Banzhaf, H. S., & Oates, W. E. (2013). On fiscal illusion in local public finance: Re-examining Ricardian equivalence and the renter effect. *National Tax Journal*, *66*(3), 511–540.

Besley, T., & Coate, S. (2003). Elected versus appointed regulators: Theory and evidence. *Journal of the European Economic Association*, *1*(5), 1176–1206.

[49] ft.com/content/9f5e7590-fec6-11e4-94c8-00144feabdc0

Besley, T., Persson, T., & Sturm, D. M., (2010). Political competition, policy and growth: Theory and evidence from the US. *Review of Economic Studies*, *77*(4), 1329–1352.
Brender, A., & Drazen, A. (2005). Political budget cycles in new versus established democracies. *Journal of Monetary Economics*, *52*(7), 1271–1295.
Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, *90*(3), 414–427.
Coggburn, J. D., & Kearney, R. C. (2010). Trouble keeping promises? An analysis of underfunding in state retiree benefits. *Public Administration Review*, *70*(1), 97–108.
Cohn, J. B., Gurun, U. G., & Moussawi, R. (2020). A project-level analysis of value creation in firms. *Financial Management*, *49*(2), 423–446.
Delli Carpini, M. X. (2000). In search of the informed citizen: What Americans know about politics and why it matters. *The Communication Review*, *4*(1), 129–164.
de Benedictis-Kessner, J., & Warshaw, C. (2016). Mayoral partisanship and municipal fiscal policy. *Journal of Politics*, *78*(4), 1124–1138.
Department of Community and Economic Development. (2015). *Municipal authorities in Pennsylvania*. Governor's Center for Local Government Services, Harrisburg, PA.
Eaton, T., & Nofsinger, J. R. (2008). Funding levels and gender in public pension plans. *Public Budgeting & Finance*, *28*(3), 108–128.
Elder, E. M., & Wagner, G. A. (2015). Political effects on pension underfunding. *Economics & Politics*, *27*(1), 1–27.
Epple, D., & Schipper, K. (1981). Municipal pension funding: A theory and some evidence. *Public Choice*, *37*(1), 141–178.
Fiorina, M. P. (1981). *Retrospective voting in American national elections*. New Haven, CT: Yale University Press.
Fischel, W. A. (2001). *The homevoter hypothesis*. Cambridge, MA: Harvard University Press.
Giertz, J. F., & Papke, L. E. (2007). "Public pension plans: Myths and realities for state budgets. *National Tax Journal*, *60*(2), 305–323.
Glaeser, E. L., & Ponzetto, G. A. M. (2014). Shrouded costs of government: The political economy of state and local public pensions. *Journal of Public Economics*, *116*(C), 89–105.
Inman, R. P. (1982). Public employee pensions and the local labor budget. *Journal of Public Economics*, *19*(1), 49–71.
Janků, J., & Libich, J. (2019). Ignorance isn't bliss: Uninformed voters drive budget cycles. *Journal of Public Economics*, *173*, 21–43.
Kramer, G. H. (1971). Short-term fluctuations in US voting behavior, 1896–1964. *American Political Science Review*, *65*(1), 131–43.
Lim, C. S. H. (2013). Preferences and incentives of appointed and elected public officials: Evidence from state trial court judges. *American Economic Review*, *103*(4), 1360–1397.
Lizzeri, A., & Persico, N. (2005). A drawback of electoral competition. *Journal of the European Economic Association*, *3*(6), 1318–1348.
Mumy, G. E. (1978). The economics of local government pensions and pension funding. *Journal of Political Economy*, *86*(3), 517–527.
Munnell, A. H., Aubry, J., & Quinby, L. (2010). *Public pension funding in practice*. NBER Working Paper No. 16442. National Bureau of Economic Research, Cambridge, MA.
Nordhaus, W. D. (1975). The political business cycle. *Review of Economic Studies*, *42*(2), 169–190.
Novy-Marx, R., & Rauh, J. (2011). Public pension promises: How big are they and what are they worth? *Journal of Finance*. *66*(4), 1211–1249.
Ornaghi, A. (2019). Civil service reforms: Evidence from U.S. police departments. Working paper. https://www.ariannaornaghi.com/research.
Pennsylvania Public Employee Retirement Commission. Various years. *Status report on local government pension plans*. Public Employee Retirement Commission, Harrisburg, PA.
Persson, T., & Tabellini, G. (2002). Political economics and public finance. In A. J. Auerbach & M. Feldstein (eds.), *Handbook of public economics* (1st ed., Vol. 3, Chapter 24, pp. 1549–1659). Philadelphia, PA: Elsevier.
Pickering, A., & Rockey, J. (2011). Ideology and the growth of government. *Review of Economics and Statistics*, *93*(3), 907–919.
Pickering, A., & Rockey, J. (2013). Ideology and the size of US state government. *Public Choice*, *156*(3-4), 443–465.
Putnam, R. D. (1995). Bowling alone, revisited. *The Responsive Community*, *5*, 18–33.
Rauh, J. D. (2017). *Hidden debt, hidden deficits: How pension promises are consuming state and local budgets*. Hoover Institution Essay, Stanford, CA.
Rogoff, K. (1990). Equilibrium political budget cycles. *American Economic Review*, *80*(1), 21–36.
Rogoff, K., & Sibert, A. (1988). Elections and macroeconomic policy cycles. *Review of Economic Studies*, *55*(1), 1–16.
Shi, M., & Svensson, J. (2006). Political budget cycles: Do they differ across countries and why? *Journal of Public Economics*, *90*(8), 1367–1389.

Smith, A., Schlozman, K. L., Verba, S., & Brady, H. (2009). *The internet and civic engagement.* Pew Internet and American Life Project.
Squire, P. (2012). *The evolution of american legislatures: Colonies, territories, and states, 1619–2009*. Ann Arbor, MI: University of Michigan Press.
Squire, P. (2017). A Squire index update. *State Politics & Policy Quarterly*, *17*(4), 361–371.
Waring, M. B. (2004a). Liability-relative investing. *Journal of Portfolio Management*, *30*(4), 8–20.
Waring, M. B. (2004b). Liability-relative investing II. *Journal of Portfolio Management*, *31*(1), 40–53.
Whalley, A. (2013). Elected versus appointed policy makers: Evidence from city treasurers. *The Journal of Law & Economics*, *56*(1), 39–81.
Wisconsin Legislative Council. Various years. *Comparative study of major public employee retirement systems.* Wisconsin Legislative Council, Madison, WI.
Wittman, D. (1983). Candidate motivation: A synthesis of alternative theories. *American Political Science Review*, *77*(1), 142–157.
Wittman, D. (1977). Candidates with policy preferences: A dynamic model. *Journal of Economic Theory*, *14*(1), 180–189.
Xu, S.-J. (2017). *Politics and hidden borrowing: Electoral cycles and state defined benefit pension plans.* Working paper. Retrieved from: https://sjxu.files.wordpress.com/2017/12/politicshiddenborrowing.pdf

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

How to cite this article: Bagchi S. The effects of political competition on the funding of public-sector pension plans. *Financial Management*. 2020;1–35. https://doi.org/10.1111/fima.12335

APPENDIX A: DETAILS OF THE THEORETICAL MODEL

The Appendix lays out the theoretical model and provides the key comparative statics result that predicts a negative relationship between political competition and the funding level of a public pension plan. The first subsection (A.1) offers an economic model by which workers allocate consumption across the two periods of this model while the second subsection (A.2) lays out the process by which election promises are made and voting occurs. The model builds closely on Besley, Persson, and Sturm (2010) (henceforth BPS) but differs in an important respect: In our model politicians facing more intense political competition are likely to promise (and enact) less responsible pension plan funding policies, whereas in BPS, politicians facing a more politically competitive environment enact more pro-growth policies. What may explain the difference between these models is that while BPS focus on the choice of policies that are easily visible to voters, this paper focuses on the effects of political competition on a policy – the funding level of a public pension – that has historically not been visible to the average voter.

The model also relies on Glaeser and Ponzetto (2014) which establishes that if some voters are less informed about the generosity of pension benefits promised by politicians to public-sector employees than the latter are, politicians seeking to be elected will use pension promises to gain votes among informed public-sector workers. While the authors take the fraction of pension benefits that is pre-funded as a parameter exogenous to their model and endogenize the level of benefits, given the focus of this paper, the model here takes the level of benefits as exogenous and instead endogenizes the level of funding chosen by politicians based on the incentives that they face.

A.1 | The Economic Model

A.1.1 | Optimization by private-sector workers

A critical assumption of this model (and one that partly drives the differences between it and the BPS model) is that private-sector workers are not fully informed of the retirement benefits that have been promised to public-sector workers. This assumption is in line with the intuition in Epple and Schipper (1981) and the assumption made in Glaeser and Ponzetto (2014), who argue that pension obligations are shrouded because of lower availability of information about pensions than wages and because of the greater difficulty of understanding the accrual of pension obligations, in contrast to current compensation. The implication of this assumption is that private-sector workers cannot figure out whether the level of taxes announced by the parties at the start of period 1 corresponds to funding the pension plan fully or underfunding it.

I model the savings decision for the representative private-sector worker. With c_1^l and c_2^l as consumption in periods 1 and 2, her utility across both periods can be expressed as:

$$U^l(c_1^l, c_2^l) = u(c_1^l) + \beta u(c_2^l), \tag{A.1}$$

If her earnings in period 1 are W^l (given exogenously), savings in period 1 are s^l, and period 1 and period 2 taxes are T_1 and T_2 respectively, then the problem for the private-sector worker is:

$$Max_{\{s^l\}} U_{per}^l(c_1^l, c_2^l) \equiv Max_{\{s^l\}}[u(W^l - s^l - T_1) + \beta u(s^l * (1+r) - E(T_2|T_1))] \tag{A.2}$$

where the "per" subscript denotes perceived (rather than realized) utility and $E(\mathrm{T}_2|\mathrm{T}_1)$ reflects the expected value of period 2 taxes given the level of period 1 taxes. I set

$$\beta = 1/(1+r) \tag{A.3}$$

Private-sector workers do not know the level of retirement benefits offered in the public sector and are therefore unaware of whether taxes announced for period 1 are adequate to fund the pension plan fully or if they fall short of full funding. Consequently private-sector workers cannot correctly anticipate the level of taxes in period 2 that would leave the pension plan balanced at the end of that period. I make the simplifying assumption that the private-sector workers are naive and set $E(T_2|T_1) = T_1$. With that assumption, the FOC for the optimization problem in (A.2) is:

$$s^l = W^l/(2+r). \tag{A.4}$$

Period 1 taxes, T_1, must be adequate to support current compensation and partly fund the pension plan and hence:

$$T_1 = N^G * \left(W^G + a * \frac{B^G}{(1+r)} \right) \tag{A.5}$$

with a, the level of pension plan funding chosen in period 1 of the model $\epsilon[\underline{a}, \bar{a}]$ and where N^G, W^G, and B^G denote the number of employees in the public sector, and wages and pensions offered to employees in that sector. Substituting (A.4) and (A.5) in (A.2), we see that:

$$U_{per}^l = \frac{(2+r)}{(1+r)} * u\left(W^l * \frac{(1+r)}{(2+r)} - N^G \left(W^G + a * \frac{B^G}{(1+r)} \right) \right). \tag{A.6}$$

Thus the *perceived* utility of private-sector workers depends (negatively) on the product of a and B^G. Given that B^G is exogenous in the model, the perceived utility depends negatively on a.

On the other hand, the *realized* utility of private-sector workers is given by:

$$U_{act}^l = u(W^l - s^l - T_1) + 1/(1+r) * u(s^l * (1+r) - T_2) \tag{A.7}$$

Under the requirement that pension obligations must be honored in full and the constraint that the pension plan be balanced at the end of period 2, the budget balance equation for period 2 is:

$$T_2 = N^G * \left(W^G + (1-a) * B^G + \frac{B^G}{(1+r)} \right). \tag{A.8}$$

Substituting (A.4), (A.5), and (A.8) in (A.7), we can show that $\partial U^I_{act}/\partial a > 0$. Thus, the realized utility of private-sector workers depends positively on a, the pension plan funding level chosen in period 1. This discussion illustrates a key feature of the model that while private-sector voters vote based on the utility they anticipate relying on the promises made (and hence prefer a *lower* pension plan funding level as that corresponds to lower taxes), their realized utility is higher if a *higher* funding level is chosen in the first period of the model as it allows for a smoother consumption path.

A.1.2 | Optimization by public-sector workers

Unlike private-sector workers, public-sector workers are aware of the benefits that they have been promised and will receive in the future. Thus the decision by either party to set a low level of taxes in period 1 (and underfund the pension plan) does not influence how they vote. The optimization problem for a representative public-sector worker is:

$$Max_{\{s^G\}} U(c_1^G, c_2^G) = Max_{\{s^G\}} \left[u(W^G - s^G - T_1) + \frac{1}{(1+r)} u(s^G * (1+r) + B^G - T_2) \right], \tag{A.9}$$

where T_1 is given by (A.5). Public-sector workers know three elements of (A.5): N^G, W^G, and B^G and the platforms announced by the parties. Thus, using (A.5), they can infer a, the funding level chosen for the pension plan in period 1. Accordingly, public-sector workers can infer T_2 using (A.8). Thus when $a < 1$ such a worker will correctly anticipate the increased taxes in period 2 and adjust her savings behavior accordingly. As a result she saves optimally and because her utility (perceived and actual) do not depend on a, she would not condition her voting behavior on a.[50]

A.2 | The Political Model

A.2.1 | Voters

Voters are heterogeneous along two dimensions, the first being the level of partisanship and the second being the sector in which they work. We assume that a fraction of the voters, $(1-\sigma)$ are partisans who prefer their respective parties because of ideological considerations and the utility gain for these voters from having their preferred party in office overrides any economic concerns. Of these partisan voters, a fraction $\frac{(1+\lambda)}{2}$ favor the Democratic party. Without loss of generality, we assume for the purposes of this discussion that the Democratic party has the edge among partisan voters and accordingly $\lambda > 0$. We characterize the remaining voters as "swing" and distinguish between them based on their sector. A fraction of these swing voters, α, work in the public sector, while the remaining $(1-\alpha)$ work in the private sector. We take the fraction of partisan voters $(1-\sigma)$, the fraction of such voters supporting the Democratic party $(\frac{(1+\lambda)}{2})$, and the fraction of swing voters who work in the public-sector (α) as exogenous parameters of the model.

For the swing voters who vote based on a combination of economic and ideological considerations, let v^{ij}_{per} and v^{ij}_{act} denote the utility of voter i belonging to group j as perceived by the voters prior to voting and as realized after the implementation of policies respectively, $j \epsilon \{I, G\}$. Voting behavior depends on v^{ij}_{per}; actual realized voter well-being depends on v^{ij}_{act}. U^{ij}_{per} and U^{ij}_{act} capture the perceived and actual (realized) economic well-being of voter i belonging to group j respectively. In general, $U^{iI}_{per} \neq U^{iI}_{act}$ whereas $U^{iG}_{per} = U^{iG}_{act}$ for the reasons provided earlier.

[50] I assume that benefits are paid out in all states of the world *regardless of funding levels*. I do so because I am not aware of a *single* case in which a Pennsylvania municipality defaulted on its pension promises. Even municipalities facing fiscal stress (such as Scranton which in 2012 paid all its employees the minimum wage because they "ran out of money") did not default on their pension commitments. As a result, the assumption that public-sector workers do not care about the level of funding in the pension plan is reasonable, at least for the Pennsylvania local governments analyzed in this paper. https://www.npr.org/2012/07/07/156416876/scrantons-public-workers-pay-cut-to-minimum-wage

A.2.2 | Parties

Reflecting the two-party system in the U.S. and the fact that municipal elections in the state of Pennsylvania are held on partisan lines, we model two parties – the Democrats and Republicans. I let the utility of politicians from each party be a function of both the ego rents they obtain from winning office and the *realized* utility of all voters, which depends on policies chosen. Voters decide who to vote for based on the utility they perceive on the basis of policies announced by parties whereas actual voter utility is realized only after the fact, i.e. after policies have been implemented. Because of the nature of the fiscal illusion, some policies that may initially appear favorable to voters (and hence popular with them) may harm them in the long run.

Let P_D denote the probability that the Democratic party wins when the level of period 1 taxes announced by the two parties are T_{1D} and T_{1R} respectively. Given that there is a one-to-one mapping between the level of period 1 taxes announced, T_{1p} and the level of funding chosen, a_p, via (A.5), we can express the probability of winning in terms of the level of funding implicitly chosen, $p\epsilon\{D,R\}$. Let E denote the ego rents for politicians from coming to office.[51] Thus, I express V_p, the utility of politician from party p, as:

$$V_p = P_p(a_p, a_{-p}) * \left[E + \sum_j N^j U^j_{act}(a_p)\right] + (1 - P_p(a_p, a_{-p})) * \sum_j N^j U^j_{act}(a_{-p}) \tag{A.10}$$

where $U^j_{act}(a_p)$ is the ex-post economic utility realized for voters belonging to group j when a_p is the level of pension plan funding chosen. More simply, V_p is given by:[52]

$$V_p = P_p(a_p, a_{-p}) * [E + \sum_j N^j(U^j_{act}(a_p) - U^j_{act}(a_{-p}))] + \sum_j N^j U^j_{act}(a_{-p}) \tag{A.11}$$

I express the perceived utility of voters based on the policy chosen as: $v^{ij}_{per}(a_p) = U^j_{per}(a_p) + (\omega^{ij} + \eta) * \mathcal{I}_D$, where $\mathcal{I}_{\mathcal{D}}$ takes a value of unity if the Democratic party wins the election and zero otherwise. Here ω^{ij} is an individual-specific parameter in favor of (or against) the Democratic party and η is a random variable capturing the aggregate popularity shock of the Democratic party in the whole population. Individuals with $\omega^{ij} > 0$ (< 0) have a bias in favor of (or against) the Democratic party, which is stronger the greater ω^{ij} is (in absolute value). I assume that $\omega^{ij} \sim U[-\frac{1}{2\phi}, \frac{1}{2\phi}]$. Thus each group has members inherently biased towards each of the parties.[53] Lastly, I assume that $\eta \sim U[-\frac{1}{2\xi}, \frac{1}{2\xi}]$. The specific realization of η is unknown to the parties when they announce their platforms, making the election outcome uncertain.

Given the above set-up and the distributional assumption on ω^{ij}, the number of voters of each type voting for the two parties can be summarized in the form of the following table:

Voter Type	Number of voters	Fraction supporting the Democratic party	Fraction supporting the Republican party
Partisan	$(1-\sigma)$	$\frac{(1+\lambda)}{2}$	$\frac{(1-\lambda)}{2}$
Public-sector swing	$\sigma * \alpha$	$1/2 + \phi * \eta$	$1/2 - \phi * \eta$
Private-sector swing	$\sigma * (1-\alpha)$	$1/2 + \phi * (\eta + U^l_{per}(a_D) - U^l_{per}(a_R))$	$1/2 - \phi * (\eta + U^l_{per}(a_D) - U^l_{per}(a_R))$
Total	1		

[51] For simplicity, I let the ego rents, E, be the same for politicians of both parties.

[52] In order to rule out the possibility that the candidate wants to lose the election in order to maximize his utility, I impose the constraint that E, the ego rents from office, are large enough such that $[E + \sum_j N^j(U^j_{act}(a_p) - U^j_{act}(a_{-p}))] > 0$ for all possible choices of a_p and a_{-p}, $p\epsilon\{D,R\}$ and $j\epsilon\{I,G\}$.

[53] In an earlier version of the paper, the distribution of party bias was allowed to vary across groups. Likewise, groups were allowed to differ in the extent to which they cared about ideology relative to economic well-being. Extending the model along those dimensions is straight-forward but because they do not appear to offer any additional insights, the presentation has been simplified by assuming that (i) the distribution of party bias and (ii) the relative weight placed on ideology relative to economic well-being is identical across public- and private-sector workers.

A Democratic victory occurs when votes received by Democrats > votes received by Republicans, the probability of which $= P[\{(1-\sigma) * \frac{(1+\lambda)}{2} + \sigma * \alpha * (1/2 + \phi * \eta) + \sigma * (1-\alpha) * (1/2 + \phi * (\eta + U^l_{per}(a_D) - U^l_{per}(a_R)))\} > \{(1-\sigma) * \frac{(1-\lambda)}{2} + \sigma * \alpha * (1/2 - \phi * \eta) + \sigma * (1-\alpha) * (1/2 - \phi * (\eta + U^l_{per}(a_D) - U^l_{per}(a_R)))\}]$

Because $\eta \sim U[-\frac{1}{2\xi}, \frac{1}{2\xi}]$, the probability of a Democratic victory simplifies to:

$$P_D(a_D, a_R, \varkappa) = \begin{cases} 1 \text{ if } \xi[(1-\alpha) * (U^l_{per}(a_D) - U^l_{per}(a_R)) - \varkappa] \geq 1/2 \\ 0 \text{ if } \xi[(1-\alpha) * (U^l_{per}(a_D) - U^l_{per}(a_R)) - \varkappa] \leq -1/2 \\ 1/2 + \xi[(1-\alpha) * (U^l_{per}(a_D) - U^l_{per}(a_R)) - \varkappa] \text{ otherwise} \end{cases}$$

where $\varkappa = \frac{-(1-\sigma)*\lambda}{2*\sigma*\phi}$. Thus the model predicts that the Democrats' chances of electoral success depend on the (endogenous) utility difference perceived on the basis of promises made $(U^l_{per}(a_D) - U^l_{per}(a_R))$, and the (exogenous) electoral advantage parameter $\varkappa$.

Drilling deeper into the parameter $\varkappa$ offers insights into the factors that make political competition more intense, which corresponds to values of $\varkappa$ closer to zero. Political competition increases as σ or the proportion of swing (partisan) voters increases (decreases). Political competition also increases as λ approaches zero, i.e. when the advantage that either party enjoys among partisan voters declines. Lastly, an increase in ϕ, which results in a more ideologically neutral set of swing voters, raises political competition.[54] Therefore, while I subsequently present a single comparative statics result, viz. the effect of an increase in this measure of political competition on the equilibrium pension plan funding level chosen, that result subsumes three different channels through which political competition may increase – either an increase in the proportion of swing voters or a decline in the edge enjoyed by the leading party among partisan voters or an increase in the degree to which swing voters are swayed by economic considerations as compared to ideology.

Substituting the above expression for $P_D(a_D, a_R, \varkappa)$ in (A.11), now shows that the expected payoff enjoyed by politicians from the Democratic party,

$$V_D = [1/2 + \xi[(1-\alpha) * (U^l_{per}(a_D) - U^l_{per}(a_R)) - \varkappa]] * \left[E + \sum_j N^j(U^j_{act}(a_D) - U^j_{act}(a_R))\right] + \sum_j N^j U^j_{act}(a_R) \tag{A.12}$$

assuming an interior solution for P_D. Given that in (A.1.2) we show that the welfare of public-sector workers is unaffected by the level of funding chosen and by definition, partisanship trumps economic considerations for partisan voters, the above expression simplifies to:

$$V_D = [1/2 + \xi[(1-\alpha) * (U^l_{per}(a_D) - U^l_{per}(a_R)) - \varkappa]] * [E + \sigma * (1-\alpha) * (U^l_{act}(a_D) - U^l_{act}(a_R))] + \sum_j N^j U^j_{act}(a_R) \tag{A.13}$$

This expression hints at the trade-offs faced by a politician from the Democratic party: While promising a lower level of funding increases his chances of winning (through its effect on $U^l_{per}(a_D)$), it also has the effect of lowering utility for private-sector workers (through its effect on $U^l_{act}(a_D)$) as lower pension plan funding leads to a sub-optimal consumption path for these workers. Politicians of both parties decide on an optimal level of funding balancing these tradeoffs, taking as given the degree of political competition measured by $\varkappa$.

[54] The last result can be seen more readily in case ω^{ij} had a smooth unimodal distribution. In that case, a shift of the mass in the distribution towards the middle would raise the probability distribution function, g_ω in that range. An increase in the density ϕ of the assumed uniform distribution could be thought of as approximating such a shift towards a more ideologically neutral electorate.

A.3 | Equilibrium

An equilibrium of the model can be represented by a pair of funding levels $\{a_D, a_R\}\epsilon[\underline{a}, \bar{a}]$, which forms a Nash equilibrium in the pre-election game between the two parties, given the equilibrium behavior of voters. We study an equilibrium where the following conditions hold:

(1) $\partial U^l_{act}(\bar{a})/\partial a = 0$ and
(2) $\xi * (1-\alpha) * (\partial U^l_{per}(\underline{a})/\partial a) * E + 1/2 * \sigma * (1-\alpha) * \partial U^l_{act}(\underline{a})/\partial a > 0$

Intuitively, the first condition requires that private-sector voter utility is being maximized at $a = \bar{a}$ while the second condition requires that the marginal benefit to private-sector workers of increasing pension plan funding levels at the point of maximum distortion, $a = \underline{a}$ is large enough to dominate the marginal cost to the politician of increasing funding levels at that point. Under these conditions, an equilibrium exists and the effect of political competition on pension plan funding level has three ranges:

1. For political competition, $\varkappa$, below a lower threshold ($\varkappa_L$), the Democrats win for sure and pursue a responsible pension plan funding policy and set $a_D = \bar{a}$.[55]
2. For $\varkappa$ in an intermediate range, the Republicans pick lower (and less responsible funding policies) than the Democrats. As competition increases, the probability that the Republicans win increases and the Democrats also move towards a policy of lower pension plan funding.
3. For $\varkappa$ close enough to zero, the party ranking and the effect of political competition on pension plan funding level are ambiguous.

There are two take-aways from these propositions. First is the result central to this paper that an increase in political competition is likely to result in a decline in pension plan funding levels. That is the key prediction of the model and one I take to the data. The second result is that any observed effects of partisanship are likely to result from variations in the intensity of political competition, rather than from the identity of the incumbent party. Subsection 4.2.3 examines this issue and obtains results that support this prediction.

A.4 | Proofs

Lemma A1. An equilibrium exists.

Proof: We show the existence of an equilibrium by taking the choice of pension plan funding by the Republican party, a_R, as given and then establishing that there is a value of $a_D\epsilon[\underline{a}, \bar{a}]$ that constitutes a best response on the part of the Democratic party. Consider the derivative of the expected payoff function for the Democratic party with respect to its pension plan funding level:

$H(a_D) = \xi(1-\alpha) * \partial U^l_{per}(a_D)/\partial a * [E + \sigma * (1-\alpha) * (U^l_{act}(a_D) - U^l_{act}(a_R))] + [1/2 + \xi[(1-\alpha) * (U^l_{per}(a_D) - U^l_{per}(a_R)) - \varkappa]] * \sigma * (1-\alpha) * \partial U^l_{act}(a_D)/\partial a$

Evaluate $H(a_D)$ at $a_D = \underline{a}$.

$H(\underline{a}) = \xi(1-\alpha) * \partial U^l_{per}(\underline{a})/\partial a * [E + \sigma * (1-\alpha) * (U^l_{act}(\underline{a}) - U^l_{act}(a_R))] + [1/2 + \xi[(1-\alpha) * (U^l_{per}(\underline{a}) - U^l_{per}(a_R)) - \varkappa]] * \sigma * (1-\alpha) * \partial U^l_{act}(\underline{a})/\partial a$

Given that $\partial U^l_{per}(\underline{a})/\partial a < 0$ and $(U^l_{act}(\underline{a}) - U^l_{act}(a_R)) \le 0$, $(\partial U^l_{per}(\underline{a})/\partial a) * (U^l_{act}(\underline{a}) - U^l_{act}(a_R)) \ge 0$. Likewise, because $(U^l_{per}(\underline{a}) - U^l_{per}(a_R)) \ge 0$ and $\partial U^l_{act}(\underline{a})/\partial a > 0$, $(U^l_{per}(\underline{a}) - U^l_{per}(a_R)) * \partial U^l_{act}(\underline{a})/\partial a \ge 0$. Accordingly, $H(\underline{a}) \ge \xi(1-\alpha) * \partial U^l_{per}(\underline{a})/\partial a * E + [1/2 - \xi\varkappa] * \sigma * (1-\alpha) * \partial U^l_{act}(\underline{a})/\partial a$

Or, $H(\underline{a}) \ge \xi(1-\alpha) * \partial U^l_{per}(\underline{a})/\partial a * E + 1/2 * \sigma * (1-\alpha) * \partial U^l_{act}(\underline{a})/\partial a > 0$ per assumption 2. Thus, $a_D > \underline{a}$.

Now we evaluate $H(a_D)$ at $a_D = \bar{a}$. Given that, $\partial U^l_{act}(\bar{a})/\partial a = 0$, per assumption 1,

$H(\bar{a}) = \xi(1-\alpha) * \partial U^l_{per}(\bar{a})/\partial a * [E + \sigma * (1-\alpha) * (U^l_{act}(\bar{a}) - U^l_{act}(a_R))] < 0.$

[55] Recall that for ease of presentation, the Democrats have a surplus of committed voters.

Because $H(a_D)$ at $a_D = \bar{a} < 0$, we can say that $a_D < \bar{a}$. Because $H(.)$ is continuous, there exists (by the intermediate value theorem) an $a_D^* \epsilon [\underline{a}, \bar{a}]$ such that $H(a_D^*) = 0$. Accordingly a_D^* constitutes a best response to a given pension plan funding level by the Republican party, a_R and hence an equilibrium exists.

Define $-\varkappa^L = \frac{1}{2\xi} + (1-\alpha) * (U_{per}^l(\underline{a}) - U_{per}^l(\bar{a}))$.

Lemma A2. If $\varkappa \leq \varkappa^L$ the Democratic Party wins for sure and picks $a_D^* = \bar{a}$.

Proof: This follows by observing that for $\varkappa \leq \varkappa^L$, the Democrats win for sure. Accordingly they maximize their utility by choosing the most responsible funding level which maximizes the utility of voters in the private-sector, i.e. $a_D^* = \bar{a}$.

Now define:

$$-\varkappa_H = -\varkappa_L + \frac{\frac{\partial U_{per}^l(\underline{a})}{\partial a}}{\sigma * \frac{\partial U_{act}^l(\underline{a})}{\partial a}} * [E + (\sigma * (1-\alpha) * (U_{act}^l(\underline{a}) - U_{act}^l(\bar{a})))]$$

Lemma A3. For $\varkappa \epsilon [\varkappa_L, \varkappa_H]$, $a_R^* = \underline{a} < a_D^* < \bar{a}$.

Proof: First we show that for all $\varkappa < \varkappa_H$, the Republicans will pick $a_R^* = \underline{a}$. To prove this it would suffice to prove that the derivative of the expected payoff for the Republican party at $a_R^* = \underline{a} < 0$.

The derivative of the expected payoff for the Republican party at $a_R^* = \underline{a}$ and $a_D^* = \bar{a}$ is:

$G(a_R) = 1/2 * \sigma * (1-\alpha) * \partial U_{act}^l(\underline{a})/\partial a + \xi(1-\alpha) * \partial U_{per}^l(\underline{a})/\partial a * [E + \sigma * (1-\alpha) * (U_{act}^l(\underline{a}) - U_{act}^l(\bar{a}))] - \xi(1-\alpha) * (U_{per}^l(\bar{a}) - U_{per}^l(\underline{a})) * \sigma * (1-\alpha) * \partial U_{act}^l(\underline{a})/\partial a + \xi\varkappa * \sigma * (1-\alpha) * \partial U_{act}^l(\underline{a})/\partial a$

Or, $G(a_R) < 1/2 * \sigma * (1-\alpha) * \partial U_{act}^l(\underline{a})/\partial a + \xi(1-\alpha) * \partial U_{per}^l(\underline{a})/\partial a * [E + \sigma * (1-\alpha) * (U_{act}^l(\underline{a}) - U_{act}^l(\bar{a}))] - \xi(1-\alpha) * (U_{per}^l(\bar{a}) - U_{per}^l(\underline{a})) * \sigma * (1-\alpha) * \partial U_{act}^l(\underline{a})/\partial a + \xi\varkappa_H * \sigma * (1-\alpha) * \partial U_{act}^l(\underline{a})/\partial a$. But because of the definitions of $\varkappa_L$ and $\varkappa_H$, this expression simplifies to 0. Thus, the derivative of the expected payoff for the Republican party at $a_R^* = \underline{a}$ and $a_D^* = \bar{a}$ is less than zero, suggesting that for all $\varkappa < \varkappa_H$, $a_R^* = \underline{a}$. Moreover, this holds for all $a_D > \underline{a}$.

Second, we show that it is optimal for the Democrats to pick $a_D^* > \underline{a}$. If not, suppose $a_D^* = \underline{a}$, bearing in mind that $a_R^* = \underline{a}$. In that case,

$H(\underline{a}) = \xi(1-\alpha) * \partial U_{per}^l(\underline{a})/\partial a * E + [1/2 - \xi\varkappa] * \sigma * (1-\alpha) * \partial U_{act}^l(\underline{a})/\partial a$

Or, $H(\underline{a}) > \xi(1-\alpha) * \partial U_{per}^l(\underline{a})/\partial a * E + 1/2 * \sigma * (1-\alpha) * \partial U_{act}^l(\underline{a})/\partial a$, given that $\varkappa < 0$.

But by virtue of assumption 2, the expression on the right hand side is positive and accordingly $H(\underline{a}) > 0$, suggesting that $a_D^* > \underline{a}$.

Likewise, $H(\bar{a}) = \xi(1-\alpha) * \partial U_{per}^l(\bar{a})/\partial a * [E + \sigma * (1-\alpha) * (U_{act}^l(\bar{a}) - U_{act}^l(\underline{a}))] < 0$ suggesting that $a_D^* < \bar{a}$.

Lastly, we observe that $a_D(a_R = \underline{a})$ is defined from:

$H(a_D) = \xi(1-\alpha) * \partial U_{per}^l(a_D)/\partial a * [E + \sigma * (1-\alpha) * (U_{act}^l(a_D) - U_{act}^l(\underline{a}))] + [1/2 + \xi[(1-\alpha) * (U_{per}^l(a_D) - U_{per}^l(\underline{a})) - \varkappa]] * \sigma * (1-\alpha) * \partial U_{act}^l(a_D)/\partial a = 0$

Using the implicit function theorem and exploiting the concavity of U_{act}^l and U_{per}^l and the fact that $\partial U_{act}^l(a_D)/\partial a$ and $\partial U_{per}^l(a_D)/\partial a$ have opposite signs, we can show that $\frac{\partial a_D^*}{\partial \varkappa} < 0$. Thus as political competition increases, the funding level promised by the Democrats in equilibrium goes down.

Lemma A.4: There exists $\varkappa > \varkappa_H$, for which we have an interior equilibrium with $a_p \epsilon [\underline{a}, \bar{a}]$ for $p \epsilon \{D, R\}$.

Proof. For $\varkappa = 0$, assumption 2 implies that both parties will pick $a_p > \underline{a}$ for $p \epsilon \{D, R\}$. Moreover, as strategies are continuous in $\varkappa$, this holds for some $\varkappa < 0$. Therefore, we may arrive at a point at which an increase in political competition no longer results in a decline in pension plan funding.

APPENDIX B: CALIBRATING ALL REPORTED FUNDED RATIOS TO A COMMON INTEREST RATE

In the paper I use the funded ratios as reported by municipalities themselves in calculating the effects of political competition. The implicit assumption I make in doing so is that the interest rate used by municipalities in discounting their plan liabilities is uncorrelated with their underlying levels of political competition. An earlier version of this paper (Bagchi 2016) suggests that this is not the case; instead, it appears that municipalities that are more competitive also choose higher interest rates.[56] That finding strengthens the conclusions with respect to the effects of political competition and suggests that our estimates may be lower bounds of the true effects of political competition on the funding level of public pensions.

Nevertheless, one can attempt to recalculate the reported funded ratios on the basis of a common interest rate. The task is challenging because pension plans rarely disclose the stream of cash flows that are discounted to arrive at an estimate of the actuarial liabilities. One has to go through an elaborate series of calculations to "reverse-engineer" the underlying cash flows before discounting them back and arriving at estimates of the liabilities for various different interest rates (Novy-Marx and Rauh 2011). In this case, however, not all of the data that are necessary for undertaking the series of steps are available making it impossible to replicate that process.[57] Beyond that limitation, data regarding the interest rates chosen by the various plans are absent from the biennial reports for 1985–2017 and are available only over the period from 2003–2013.

I deal with these data limitations by, first, noting that estimates of the effective average duration of pension liabilities range from 11 years (Rauh 2017) to 13 years (Novy-Marx and Rauh 2011) and even to 15 years (Waring 2004a, 2004b). Second, lacking data on interest rates for each year for which the data on funded ratios are available, I assume that the interest rate used by the plans for the year 2003[58] is what was used over the entire sample period from 1985–2017. Although this assumption is almost certainly not likely to hold, it is relatively innocuous. Examining the period between 2003–2013, we see that interest rates used by plans are not very sensitive to broader macro-economic trends. For example, the median interest rate used by all local plans during this period stayed at 7 percent all throughout, notwithstanding the large swings in the Federal funds rate which ranged between 0.12 percent (in 2009) and 5.26 percent (in 2007). Moreover, 2003 is reasonably close to the mid-point of the overall sample period, 1985–2017. I discount all liabilities with respect to 7 percent, corresponding to the median across all municipal pension plans in the sample.[59] Table A.3 presents the results for three choices of the weighted average duration of liabilities – 11 years, 13 years, and 15 years with liabilities for all plans discounted to 7 percent.

As we can see, the coefficients on political competition continue to be negative and statistically significant all throughout for municipal plans. In terms of magnitude, a one standard deviation increase in the level of political competition is associated with an approximately 3.5 percent decline in the funded ratio. As was the case earlier, there is no relationship between the level of political competition and the funding status for pension plans run by municipal authorities.

As the 7 percent rate is likely too high, I also discount them back to an interest rate corresponding to the nominal yield on zero-coupon Treasury bonds of similar duration. Based on recent market conditions and expectations of market participants about future economic conditions, I use 1.5 percent for the real yield on long-term zero-coupon Treasury bonds and add in 2 percent to reflect inflation expectations, for a nominal yield of 3.5 percent. I continue to find a negative and significant relationship between political competition and the funded ratio for municipal plans, with no such effects for municipal authority plans. These results are available on request.

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[56] I estimate the effect of a one standard deviation increase in the level of political competition is to increase the rate used for discounting actuarial liabilities by about 4–6 basis points. Likewise, I also find a positive relationship between political competition and the amortization period chosen to pay off unfunded liabilities with an increase in political competition by 1 standard deviation driving an increase in the amortization period by about 0.5 years. As before, there is no relationship between political competition and the interest rate or amortization period for plans run by municipal authorities. These results are in line with prior work which shows that states make assumptions strategically when faced with fiscal stress (Giertz and Papke, 2007) and they are available on request.

[57] For example, data on the benefit factor which captures the added benefit available for an additional year of service or data on the nature of COLA adjustments in these plans are unavailable in these datasets.

[58] In about 1 percent of cases where data on interest rates are unavailable for 2003, I use data from a subsequent year.

[59] Consider a plan which reports liabilities of \$130,000 and assets of \$125,000 per member and discounts liabilities at an interest rate of 7.5 percent. If I were to recalibrate this plan's reported funded ratio (96.2 percent) to an interest rate of 7 percent, then assuming that the weighted average duration of liabilities is 13 years, the true liabilities for this plan = \$130,000 * (1.075/1.07)^13 = \$138,122. Accordingly the recalibrated funded ratio = 125000/138122 or 90.5 percent.

The Effects of Political Competition on the Funding of Public-Sector Pension Plans

August 22, 2020

Internet Appendix[1]

Sutirtha Bagchi[2]

[1]I thank Sharon Bogden of the Department of General Services of Pennsylvania, Bernard Kozlowski of the Pennsylvania Public Employee Retirement Commission, Rhonda R. Newton of the Pennsylvania Heritage Foundation, and James M. Snyder Jr. for help in obtaining data. I also thank Subhayu Bandyopadhyay, Sourav Bhattacharya, Tilman Borgers, Richard Boylan, Sebastien J. Bradley, Spencer T. Brien, Charles C. Brown, Traviss Cassidy, Julie Berry Cullen, David J. Denis, Raymond Fisman, Kyle Handley, James R. Hines Jr., Jeffrey L. Hoopes, Lawrence W. Kenny, Erasmus Kersting, Christopher Kilby, Manoj Kulchania, Scott Masten, Rabih Moussawi, Dana M. Muir, Michael Pagano, Leslie E. Papke, Giacomo Ponzetto, Stephen Salant, Nathan Seegert, Dan Silverman, Jagadeesh Sivadasan, Joel Slemrod, Mehmet S. Tosun, Ugo Troiano, Rodrigo Wagner, and seminar participants at the University of Michigan, International Institute of Public Finance Annual Congress, and National Tax Association Conference for helpful comments. Data acquisition for the project was made possible in part by grants from the Office of Tax Policy Research, the Doctoral Studies Office at the University of Michigan, and Villanova University. All links were active as of 06/10/2020. All errors remain my own.

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Table A.1: Effect of Political Competition on Actuarial Funded Ratio, Unfunded Liabilities (variously normalized) for plans run by Municipal Authorities, split based on the nature of the match with municipalities

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for plans which do not have an unambiguous municipal match			Coefficient for plans which have an unambiguous match with one municipality		
Panel A: Dependent variable: Actuarial Funded Ratio						
Political Competition	7.479	8.945	9.240	-27.52	-39.35	-38.52
	(54.89)	(57.65)	(58.35)	(46.27)	(43.44)	(43.28)
Number of observations	415	415	415	432	432	432
R^2	0.67	0.69	0.69	0.66	0.66	0.66
Panel B: Dependent variable: Unfunded liabilities per active member						
Political Competition	-32.86	-33.49	-33.83	-11.26	-13.30	-13.31
	(37.29)	(37.93)	(37.55)	(33.79)	(33.33)	(33.24)
Number of observations	408	408	408	416	416	416
R^2	0.70	0.70	0.70	0.57	0.57	0.57
Panel C: Dependent variable: Unfunded liabilities as a percent of payroll						
Political Competition	-36.17	-36.18	-36.90	-41.69	-43.61	-43.34
	(102.8)	(104.1)	(102.9)	(91.32)	(90.37)	(89.44)
Number of observations	399	399	399	405	405	405
R^2	0.71	0.72	0.72	0.58	0.58	0.58
Panel D: Dependent variable: Unfunded liabilities per capita						
Political Competition	-61.90	-70.14	-72.75	19.33	18.82	18.74
	(92.01)	(96.14)	(93.99)	(36.64)	(40.89)	(41.07)
Number of observations	408	408	408	416	416	416
R^2	0.61	0.61	0.61	0.65	0.65	0.66
Municipality and period FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Municipal demographic controls		✔	✔		✔	✔
Municipal fiscal control			✔			✔

Robust standard errors clustered at the municipality level are in parentheses; * p < 0.10, ** p < 0.05, *** p < 0.01.
Regressions estimated on all defined benefit pension plans from Pennsylvania run by municipal authorities for the period 1985–2017. Unfunded liabilities per active member have been rescaled by dividing by $1,000 for ease of presentation. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$. Controls for class of employee covered, municipality and period fixed effects are included in all specifications. Municipal demographic controls included in columns (2), (3), (5), and (6) are the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the local unemployment rate, and the log of per capita income. Municipal fiscal control included in columns (3) and (6) is the ratio of debt to taxes collected by the municipality. The dependent variables and all control variables have been winsorized at the 5 and 95% levels.

Table A.2: Effect of Political Competition on Unfunded Liabilities and Log of Unfunded Liabilities variously normalized

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Panel A: Dependent variable: Unfunded liabilities per active member						
Political Competition	48.12***	48.54***	48.53***	-25.61	-28.23	-28.46
	(11.47)	(11.56)	(11.57)	(25.36)	(25.16)	(25.09)
Number of observations	5509	5509	5509	824	824	824
R^2	0.56	0.56	0.56	0.66	0.67	0.67
Panel B: Dependent variable: Unfunded liabilities as percent of payroll						
Political Competition	94.33***	100.5***	100.5***	-45.77	-56.02	-56.75
	(31.50)	(31.18)	(31.19)	(68.14)	(66.90)	(66.58)
Number of observations	5457	5457	5457	804	804	804
R^2	0.60	0.60	0.60	0.69	0.69	0.69
Panel C: Dependent variable: Unfunded liabilities per capita						
Political Competition	70.75***	74.98***	74.96***	-30.66	-38.81	-40.55
	(21.75)	(21.80)	(21.80)	(54.11)	(57.58)	(56.66)
Number of observations	5509	5509	5509	824	824	824
R^2	0.56	0.57	0.57	0.61	0.61	0.61
Panel D: Dependent variable: Log of unfunded liabilities per active member						
Political Competition	1.086***	1.059***	1.060***	-0.516	-0.567	-0.567
	(0.329)	(0.331)	(0.331)	(0.367)	(0.393)	(0.390)
Number of observations	5509	5509	5509	824	824	824
R^2	0.56	0.56	0.56	0.74	0.74	0.74
Panel E: Dependent variable: Log of unfunded liabilities per capita						
Political Competition	0.170**	0.176**	0.176**	-0.0505	-0.0638	-0.0662
	(0.0799)	(0.0801)	(0.0801)	(0.0893)	(0.0926)	(0.0913)
Number of observations	5509	5509	5509	824	824	824
R^2	0.36	0.36	0.36	0.63	0.63	0.64
Municipality and period FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Municipal demographic controls		✔	✔		✔	✔
Municipal fiscal control			✔			✔

Robust standard errors clustered at the municipality level are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all public-sector defined benefit pension plans from Pennsylvania for the period 1985–2017. Unfunded liabilities per active member have been rescaled by dividing by $1,000 for ease of presentation. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$. Controls for class of employee covered, municipality and period fixed effects are included in all specifications. Municipal demographic controls included in columns (2), (3), (5), and (6) are the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the local unemployment rate, and the log of per capita income. Municipal fiscal control included in columns (3) and (6) is the ratio of debt to taxes collected by the municipality. The dependent variables and all control variables have been winsorized at the 5 and 95% levels.

Table A.3: Effect of Political Competition on Actuarial Funded Ratio, Normalized to Common Interest Rates

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Panel A: Duration of liabilities assumed = 11 years						
Political competition	-49.25***	-48.43***	-48.42***	6.173	1.427	1.257
	(13.82)	(13.55)	(13.55)	(50.82)	(50.79)	(49.98)
Number of observations	5283	5283	5283	625	625	625
R^2	0.58	0.58	0.58	0.73	0.73	0.73
Panel B: Duration of liabilities assumed = 13 years						
Political competition	-49.21***	-48.41***	-48.40***	6.251	1.468	1.290
	(13.89)	(13.61)	(13.60)	(50.78)	(50.62)	(49.77)
Number of observations	5283	5283	5283	625	625	625
R^2	0.58	0.58	0.58	0.73	0.74	0.74
Panel C: Duration of liabilities assumed = 15 years						
Political competition	-49.41***	-48.69***	-48.68***	5.969	1.241	1.051
	(13.94)	(13.65)	(13.65)	(50.57)	(50.37)	(49.47)
Number of observations	5283	5283	5283	625	625	625
R^2	0.59	0.59	0.59	0.74	0.74	0.74
Municipality and period FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Municipal demographic controls		✔	✔		✔	✔
Municipal fiscal control			✔			✔

Robust standard errors clustered at the municipality level are in parentheses; * p < 0.10, ** p < 0.05, *** p < 0.01.

Regressions estimated on all public-sector defined benefit pension plans from Pennsylvania for the period 1985–2017. The dependent variable, the actuarial funded ratio, is defined as the percent of pension liabilities funded, where the liabilities have been re-calculated assuming a 7 percent interest rate. The 7 percent discount rate used corresponds to the median interest rate across all plans. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$. Controls for class of employee covered, municipality and period fixed effects are included in all specifications. Municipal demographic controls included in columns (2), (3), (5), and (6) are the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the local unemployment rate, and the log of per capita income. Municipal fiscal control included in columns (3) and (6) is the ratio of debt to taxes collected by the municipality. The dependent variables and all control variables have been winsorized at the 5 and 95% levels.

Table A.4: Robustness Checks for the Effect of Political Competition on Actuarial Funded Ratio

	(1)	(2)	(3)	(4)	(5)	(6)
	Coefficient for municipal plans			Coefficient for plans run by municipal authorities		
Baseline Specification	-47.28***	-47.07***	-47.07***	-4.067	-1.496	-0.546
	(12.09)	(11.98)	(11.99)	(36.61)	(36.69)	(36.57)
RC1: Using average vote share	-30.26**	-26.74**	-26.77**	-6.299	3.087	2.963
based on Presidential elections	(12.44)	(12.52)	(12.53)	(37.52)	(37.30)	(36.85)
RC2: Introducing	-37.59**	-38.16**	-38.22**	4.994	1.871	2.349
county-by-period fixed effects	(15.12)	(15.01)	(15.02)	(50.89)	(50.91)	(47.85)
RC3: Using municipality-specific	-45.90**	-36.94*	-36.46*	-60.62	-61.21	-59.65
linear time trends	(18.91)	(20.63)	(20.65)	(58.18)	(61.15)	(60.59)
RC4: Controlling for the number	-33.77***	-36.40***	-36.37***	-4.616	-2.397	-2.594
of active members	(12.78)	(12.63)	(12.64)	(34.87)	(35.88)	(36.54)
RC5: Weighting regression by	-48.26***	-42.37***	-42.42***	18.52	24.36	23.47
number of active members	(12.35)	(11.52)	(11.55)	(37.47)	(37.66)	(35.59)
RC6: Only including municipal-	-47.04**	-38.66**	-38.94**	-4.067	-1.496	-0.546
ities with at least one authority	(19.85)	(19.54)	(19.69)	(36.61)	(36.69)	(36.57)
Municipality and period FE	✔	✔	✔	✔	✔	✔
Employee group dummies	✔	✔	✔	✔	✔	✔
Municipal demographic controls		✔	✔		✔	✔
Municipal fiscal control			✔			✔

Robust standard errors clustered at the municipality level are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Regressions estimated on all public-sector defined benefit pension plans from Pennsylvania for the period 1985–2017. The dependent variable, the actuarial funded ratio, is defined as the percent of pension liabilities funded. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$. Municipal demographic controls included in columns (2), (3), (5), and (6) are the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the local unemployment rate, and the log of per capita income. Municipal fiscal control included in columns (3) and (6) is the ratio of debt to taxes collected by the municipality. The dependent variables and all control variables have been winsorized at the 5 and 95% levels.

RC1 is motivated by the fact that voters may consider the performance of their local officials in casting their votes for elections to state-level offices raising the possibility of reverse causality between funding status of the pension plan and the level of political competition in the municipality. Voters are however unlikely to consider the performance of their local officials in casting their votes for the office of President. Thus, using vote share based solely on Presidential elections minimizes the possibility of reverse causality associated with using data on elections to national and state-level offices. RC4 and 5 are motivated by the fact that our regressions accord equal importance to a plan with a single member and a plan with several hundred members. RC4 addresses that issue by controlling for the log of active members whereas RC5 estimates a weighted regression, with the number of active members in the plan serving as weights. RC6 is motivated by the fact that not all municipalities which offer a pension plan also have an authority offering their own pension plan. In particular, municipalities that have a municipal authority offering their own pension plan are larger than the typical municipality offering a pension plan. Hence this robustness check limits the estimation only to pension plans run by municipalities that have at least one municipal authority.

Table A.5: Sample-split test examining the Effects of Political Competition on Funded Ratio, Based on Differences in Voter Turnout using Alternative ways of cutting the data

	(1)	(2)	(3)	(4)	(5)	(6)
	Percentage of residents who turnout to vote <= Median			Percentage of residents who turnout to vote > Median		
Panel A: Splitting the sample into periods that are 2 years long						
Political Competition	-14.73*	-10.03	-10.04	-8.318	-5.104	-4.751
	(8.598)	(8.362)	(8.342)	(7.199)	(7.303)	(7.305)
Number of observations	12673	12673	12673	12860	12860	12860
R^2	0.54	0.54	0.54	0.52	0.53	0.53
Panel B: Splitting the sample into periods that are 4 years long						
Political Competition	-28.68***	-26.41**	-26.43**	-0.130	6.636	6.875
	(10.26)	(10.26)	(10.24)	(9.029)	(8.836)	(8.862)
Number of observations	5749	5749	5749	5893	5893	5893
R^2	0.56	0.56	0.56	0.55	0.55	0.55
Panel C: Splitting the sample into periods that are 6 years long						
Political Competition	-30.52***	-22.34*	-22.44*	-3.268	-0.468	-0.280
	(11.74)	(11.59)	(11.56)	(10.65)	(10.57)	(10.55)
Number of observations	4901	4901	4901	5049	5049	5049
R^2	0.55	0.55	0.55	0.52	0.52	0.52
Panel D: Splitting the sample into periods that are 8 years long						
Political Competition	-45.96***	-36.95***	-36.80***	1.983	3.916	4.259
	(14.13)	(14.10)	(14.08)	(11.91)	(11.88)	(11.89)
Number of observations	3436	3436	3436	3479	3479	3479
R^2	0.58	0.58	0.58	0.55	0.55	0.55
Panel E: Splitting the sample into periods that are 16 years long						
Political Competition	-122.9**	-101.8**	-101.2**	-10.15	-20.08	-20.12
	(47.65)	(48.12)	(48.23)	(35.82)	(37.02)	(37.04)
Number of observations	1869	1869	1869	1882	1882	1882
R^2	0.64	0.64	0.64	0.61	0.61	0.61
Municipality and Period FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Municipal demographic controls		✔	✔		✔	✔
Municipal fiscal control			✔			✔

Robust standard errors clustered at the municipality level are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Regressions estimated on all municipal defined benefit pension plans from Pennsylvania for the period 1985–2017. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$. Data on votes cast in all national and state elections which are used in splitting the sample are sourced from successive issues of the Pennsylvania Manual and the population data come from the 1980 Decennial Census as described in fn. (**??**). Controls for class of employee covered, municipal and period fixed effects are included in all specifications. Municipal demographic controls included in columns (2), (3), (5), and (6) are the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the local unemployment rate, and the log of per capita income. Municipal fiscal control included in columns (3) and (6) is the ratio of debt to taxes collected by the municipality. The dependent variables and all control variables have been winsorized at the 5 and 95% levels.

Table A.6: Sample-split test examining the Effects of Political Competition on Funded Ratio, Based on Differences in Residential Mobility using Alternative ways of cutting the data

	(1)	(2)	(3)	(4)	(5)	(6)
	Percentage of residents who stayed in the same house <= Median			Percentage of residents who stayed in the same house>Median		
Panel A: Splitting the sample into periods that are 2 years long						
Political Competition	-13.13*	-8.588	-8.239	-5.921	-5.716	-5.733
	(7.426)	(7.461)	(7.392)	(8.956)	(8.555)	(8.549)
Number of observations	12937	12937	12937	12596	12596	12596
R^2	0.48	0.48	0.48	0.56	0.56	0.56
Panel B: Splitting the sample into periods that are 4 years long						
Political Competition	-15.27	-11.25	-10.93	-9.474	-9.240	-9.224
	(9.633)	(9.520)	(9.449)	(11.05)	(10.73)	(10.73)
Number of observations	5832	5832	5832	5810	5810	5810
R^2	0.50	0.50	0.50	0.58	0.59	0.59
Panel C: Splitting the sample into periods that are 6 years long						
Political Competition	-22.66**	-18.46	-18.06	-9.932	-5.482	-5.860
	(11.29)	(11.21)	(11.18)	(12.05)	(11.63)	(11.60)
Number of observations	5012	5012	5012	4938	4938	4938
R^2	0.48	0.48	0.49	0.56	0.57	0.57
Panel D: Splitting the sample into periods that are 8 years long						
Political Competition	-30.95**	-28.57**	-27.89**	-16.85	-8.218	-8.506
	(13.00)	(12.94)	(12.90)	(14.73)	(14.37)	(14.35)
Number of observations	3494	3494	3494	3421	3421	3421
R^2	0.51	0.51	0.51	0.60	0.60	0.60
Panel E: Splitting the sample into periods that are 16 years long						
Political Competition	-104.9***	-107.9***	-108.1***	-29.31	-10.37	-10.58
	(39.06)	(39.65)	(39.62)	(47.77)	(46.39)	(46.75)
Number of observations	1891	1891	1891	1860	1860	1860
R^2	0.57	0.57	0.57	0.65	0.65	0.65
Municipality and Period FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Municipal demographic controls		✔	✔		✔	✔
Municipal fiscal control			✔			✔

Robust standard errors clustered at the municipality level are in parentheses; * p < 0.10, ** p < 0.05, *** p < 0.01.
Regressions estimated on all municipal defined benefit pension plans from Pennsylvania for the period 1985–2017. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$. The data used in splitting the sample come from the 1980 Decennial Census as described in fn. (**??**). Controls for class of employee covered, municipal and period fixed effects are included in all specifications. Municipal demographic controls included in columns (2), (3), (5), and (6) are the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the local unemployment rate, and the log of per capita income. Municipal fiscal control included in columns (3) and (6) is the ratio of debt to taxes collected by the municipality. The dependent variables and all control variables have been winsorized at the 5 and 95% levels.

Table A.7: Sample-split test examining the Effects of Political Competition on Funded Ratio, Based on Differences in Educational Attainment using Alternative ways of cutting the data

	(1)	(2)	(3)	(4)	(5)	(6)
	Percentage of residents with a high school degree <= Median			Percentage of residents with a high school degree > Median		
Panel A: Splitting the sample into periods that are 2 years long						
Political Competition	-10.53	-7.126	-7.026	-3.521	-1.994	-1.902
	(9.157)	(8.849)	(8.841)	(7.224)	(7.219)	(7.205)
Number of observations	12605	12605	12605	12928	12928	12928
R^2	0.56	0.57	0.57	0.47	0.48	0.48
Panel B: Splitting the sample into periods that are 4 years long						
Political Competition	-20.92*	-16.62	-16.59	-6.530	-3.701	-3.629
	(10.85)	(10.65)	(10.63)	(8.913)	(8.989)	(8.996)
Number of observations	5754	5754	5754	5888	5888	5888
R^2	0.58	0.59	0.59	0.50	0.51	0.51
Panel C: Splitting the sample into periods that are 6 years long						
Political Competition	-22.59*	-15.02	-15.14	-8.482	-6.277	-6.096
	(12.43)	(12.11)	(12.06)	(10.22)	(10.30)	(10.31)
Number of observations	4893	4893	4893	5057	5057	5057
R^2	0.57	0.57	0.57	0.48	0.49	0.49
Panel D: Splitting the sample into periods that are 8 years long						
Political Competition	-48.19***	-40.76***	-40.93***	-5.440	-2.208	-1.775
	(14.84)	(14.60)	(14.55)	(11.76)	(11.78)	(11.81)
Number of observations	3402	3402	3402	3513	3513	3513
R^2	0.60	0.61	0.61	0.50	0.50	0.50
Panel E: Splitting the sample into periods that are 16 years long						
Political Competition	-80.97	-57.35	-54.41	-15.90	-19.54	-18.66
	(56.38)	(55.51)	(55.75)	(35.95)	(38.60)	(38.22)
Number of observations	1853	1853	1853	1898	1898	1898
R^2	0.66	0.66	0.66	0.56	0.56	0.56
Municipality and Period FE	✔	✔	✔	✔	✔	✔
Employee-group dummies	✔	✔	✔	✔	✔	✔
Municipal demographic controls		✔	✔		✔	✔
Municipal fiscal control			✔			✔

Robust standard errors clustered at the municipality level are in parentheses; * p < 0.10, ** p < 0.05, *** p < 0.01.
Regressions estimated on all municipal defined benefit pension plans from Pennsylvania for the period 1985–2017. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$. The data used in splitting the sample come from the 1980 Decennial Census as described in fn. (**??**). Controls for class of employee covered, municipal and period fixed effects are included in all specifications. Municipal demographic controls included in columns (2), (3), (5), and (6) are the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the local unemployment rate, and the log of per capita income. Municipal fiscal control included in columns (3) and (6) is the ratio of debt to taxes collected by the municipality. The dependent variables and all control variables have been winsorized at the 5 and 95% levels.

Table A.8: Effects of Political Competition on Municipal Debt Levels

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	D.v.: Debt per capita		D.v.: Debt/ Revenues		D.v.: Debt/ Expenses		D.v.: Debt/ Taxes	
Panel A: Measure of Political Competition: Absolute Difference of Democratic Vote Share from 50%								
Political Competition	0.230	0.244	-0.0922	-0.103	-0.102	-0.112	-0.197	-0.233
	(0.742)	(0.736)	(0.163)	(0.163)	(0.171)	(0.172)	(0.553)	(0.556)
Number of observations	5095	5095	7688	7688	7688	7688	7687	7687
R^2	0.73	0.73	0.54	0.54	0.54	0.54	0.56	0.56
Panel B: Measure of Political Competition: Absolute Difference of Democratic Vote Share from 50%, controlling for Democratic vote share								
Political Competition	0.484	0.541	-0.0592	-0.0338	-0.0650	-0.0378	0.0589	0.149
	(0.789)	(0.793)	(0.190)	(0.189)	(0.200)	(0.199)	(0.680)	(0.677)
Coefficient on Democratic vote share	-0.723	-0.886	-0.0728	-0.158	-0.0823	-0.170	-0.563	-0.870
	(0.793)	(0.878)	(0.177)	(0.185)	(0.185)	(0.193)	(0.632)	(0.662)
Number of observations	5095	5095	7688	7688	7688	7688	7687	7687
R^2	0.73	0.73	0.54	0.54	0.54	0.54	0.56	0.56
Panel C: Measure of political competition: Linear and Squared term for Democratic vote share								
Coefficient on a linear term for	1.290	1.372	-0.270	-0.257	-0.259	-0.241	-0.681	-0.638
Democratic vote share	(2.612)	(2.597)	(0.558)	(0.556)	(0.588)	(0.586)	(1.917)	(1.914)
Coefficient on a squared term	-2.034	-2.287	0.195	0.0950	0.167	0.0603	0.174	-0.181
for Democratic vote share	(2.718)	(2.730)	(0.622)	(0.615)	(0.655)	(0.648)	(2.213)	(2.199)
Number of observations	5095	5095	7688	7688	7688	7688	7687	7687
R^2	0.73	0.73	0.54	0.54	0.54	0.54	0.56	0.56
Municipality and period FE	✔	✔	✔	✔	✔	✔	✔	✔
Municipal demographic controls		✔		✔		✔		✔

Robust standard errors clustered at the county level are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
The dependent variable is log of debt per capita in columns (1) and (2), debt normalized by municipal revenues in columns (3) and (4), municipal expenses in columns (5) and (6), and municipal taxes in columns (7) and (8). The municipal demographic controls are identical to what have been used elsewhere, viz. the percentage of households that are owner-occupied, the percentage of population aged 65 or older, the local unemployment rate, and the log of per capita income. The dependent variable and all control variables have been winsorized at the 5 and 95% levels.

Table A.9: Effect of Political Competition on Actuarial Funded Ratio for State Pension Plans using Alternative ways of cutting the data

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Splitting the sample into periods that are 2 years long									
Political Competition	-14.01	-14.48	-13.94	-14.32	-14.93	-14.29			
	(9.827)	(10.19)	(10.37)	(9.657)	(9.963)	(10.26)			
Coefficient on Democratic vote share				3.603	4.603	3.723	-59.31	-59.80	-58.26
				(7.556)	(7.806)	(8.057)	(52.01)	(53.79)	(53.68)
Coefficient on Democratic vote share squared							63.48	64.91	62.44
							(50.40)	(52.07)	(53.20)
Number of observations	559	559	559	559	559	559	559	559	559
R^2	0.36	0.36	0.37	0.36	0.36	0.37	0.36	0.36	0.37
Panel B: Splitting the sample into periods that are 4 years long									
Political Competition	-20.24	-19.94	-18.01	-20.31	-19.97	-17.97			
	(17.37)	(17.71)	(17.61)	(17.53)	(17.82)	(17.63)			
Coefficient on Democratic vote share				-1.213	-0.829	0.810	-74.55	-72.14	-57.89
				(16.54)	(17.18)	(17.16)	(107.7)	(109.3)	(105.6)
Coefficient on Democratic vote share squared							74.05	71.88	59.14
							(104.6)	(106.2)	(104.5)
Number of observations	385	385	385	385	385	385	385	385	385
R^2	0.35	0.36	0.37	0.35	0.36	0.37	0.35	0.35	0.37
Panel C: Splitting the sample into periods that are 6 years long									
Political Competition	-21.15	-17.95	-13.79	-22.29	-18.93	-14.87			
	(26.76)	(26.51)	(26.89)	(27.48)	(26.83)	(27.23)			
Coefficient on Democratic vote share				-7.094	-10.28	-9.253	-88.46	-74.31	-46.92
				(20.07)	(20.55)	(19.80)	(187.4)	(184.1)	(185.0)
Coefficient on Democratic vote share squared							82.17	64.44	38.19
							(180.7)	(177.7)	(180.4)
Number of observations	244	244	244	244	244	244	244	244	244
R^2	0.38	0.38	0.39	0.38	0.38	0.39	0.37	0.38	0.39
State and Period FE	✔	✔	✔	✔	✔	✔	✔	✔	✔
Plan-specific controls		✔	✔		✔	✔		✔	✔
State socio-economic controls and fiscal control			✔			✔			✔

Robust standard errors clustered at the state level are in parentheses; * p < 0.10, ** p < 0.05, *** p < 0.01.
Regressions estimated on all state defined benefit pension plans for the period 1989–2017, which is split into periods of varying lag lengths as noted for each panel. Political competition, along with all control variables, are averages for the previous period; thus they are introduced with a lag. The dependent variable, the actuarial funded ratio, is defined as the percent of pension liabilities funded and calculated for all plans within a state. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$. State and period fixed effects are included in all specifications. Plan-specific controls include the coverage of employees under Social Security and the ratio of active members to retirees. State fiscal and socio-economic controls included are the median deviation from the state's long-term unemployment rate, the coverage of employees under collective bargaining, average state income tax rate, and the professionalization score of the state legislature. For additional details about variables and data sources, please refer to notes following Table **??**.

Table A.10: Sample-split test examining the Effects of Political Competition on Funded Ratio for State Pension Plans, Based on Differences in Voter Awareness

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Sample of state plans split using voter turnout in Presidential and Congressional elections						
	Percentage of residents who turnout to vote <= Median			Percentage of residents who turnout to vote > Median		
Political Competition	-54.63	-68.60	-54.62	-26.77	-18.81	-14.21
	(44.66)	(43.59)	(39.66)	(43.77)	(47.56)	(43.63)
Number of observations	75	75	75	71	71	71
R^2	0.62	0.64	0.69	0.39	0.41	0.44
Panel B1: Sample of state plans split using residential mobility						
	Percentage of residents who stayed in the same house <= Median			Percentage of residents who stayed in the same house > Median		
Political Competition	-83.24	-82.97	-64.73	-46.25	-42.80	-15.09
	(51.03)	(50.60)	(41.65)	(39.91)	(43.34)	(43.02)
Number of observations	74	74	74	72	72	72
R^2	0.53	0.54	0.56	0.51	0.51	0.59
Panel B2: Sample of state plans split using residential mobility						
	Percentage of residents who stayed in the same state <= Median			Percentage of residents who stayed in the same state > Median		
Political Competition	-98.83	-105.2*	-81.07	-35.74	-32.82	-9.176
	(60.36)	(58.71)	(53.89)	(36.82)	(34.72)	(33.61)
Number of observations	74	74	74	72	72	72
R^2	0.58	0.59	0.60	0.45	0.46	0.54
State and Period FE	✔	✔	✔	✔	✔	✔
Plan-specific controls		✔	✔		✔	✔
State socio-economic controls and fiscal control			✔			✔

Robust standard errors clustered at the state level are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Regressions estimated on all state defined benefit pension plans for the period 1989–2017, which is split into three time periods (1989–1998, 1999–2008, and 2009–2017) largely following the splits used for municipal plans. Political competition, along with all control variables, are averages for the previous period; thus they are introduced with a lag. The dependent variable, the actuarial funded ratio, is defined as the percent of pension liabilities funded and calculated for all plans within a state. The measure of political competition used is that defined by BPS (2010), viz. $PC_{mt} = -|0.5 - D_{mt}|$.State and period fixed effects are included in all specifications. Plan-specific controls include the coverage of employees under Social Security and the ratio of active members to retirees. State fiscal and socio-economic controls included are the median deviation from the state's long-term unemployment rate, the coverage of employees under collective bargaining, average state income tax rate, and the professionalization score of the state legislature.The dependent variable and all control variables have been winsorized at the 5 and 95% levels. For additional details about variables and data sources, please refer to notes following Table **??**.
The turnout data used for the sample-split in Panel A are sourced from McDonald (2020) and the residential mobility data used for sample-splits in Panels B1 and B2 are sourced from the 1980 Decennial Census. For Panel B1, I consider the proportion of residents who report living in the same house 5 years ago, whereas in Panel B2, I consider the proportion of residents living in the same state 5 years ago.

Table A.11: Correlation between Democratic vote share in local elections and Democratic vote share in national and state races

	(1)	(2)	(3)	(4)	(5)	(6)
	All municipalities	Smaller municipalities	Larger municipalities	All municipalities	Smaller municipalities	Larger municipalities
Panel A: Democratic vote share in local elections regressed on Democratic vote share based on elections for President and U.S. Senate						
Democratic vote share in state/ national elections	2.051***	2.029***	2.083***	1.790***	1.834***	1.767***
	(0.110)	(0.206)	(0.0981)	(0.141)	(0.254)	(0.128)
Number of observations	234	116	118	234	116	118
R^2	0.56	0.45	0.72	0.59	0.48	0.76
Panel B: Democratic vote share in local elections regressed on Democratic vote share based on elections to all down-ballot offices						
Democratic vote share in state/ national elections	1.841***	1.939***	1.770***	1.573***	1.715***	1.497***
	(0.101)	(0.192)	(0.100)	(0.134)	(0.252)	(0.109)
Number of observations	196	90	106	196	90	106
R^2	0.61	0.52	0.72	0.64	0.55	0.79
Panel C: Democratic vote share in local elections regressed on Democratic vote share based on elections for Governor						
Democratic vote share in state/ national elections	1.538***	1.438***	1.900***	1.558***	1.504***	1.760***
	(0.249)	(0.301)	(0.458)	(0.259)	(0.317)	(0.546)
Number of observations	44	30	14	44	30	14
R^2	0.37	0.33	0.57	0.38	0.35	0.59
Panel D: Democratic vote share in local elections regressed on Democratic vote share based on elections to all national and state offices						
Democratic vote share in state/ national elections	1.901***	1.951***	1.873***	1.672***	1.794***	1.598***
	(0.0963)	(0.175)	(0.0983)	(0.123)	(0.219)	(0.111)
Number of observations	240	120	120	240	120	120
R^2	0.58	0.49	0.72	0.61	0.51	0.77
Year FE	✔	✔	✔	✔	✔	✔
County FE				✔	✔	✔

Robust standard errors are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
The dependent variable in these regressions is the Democratic vote share in local elections that were held in 240 municipalities in five counties (Bucks, Chester, Dauphin, Lancaster, and Lehigh) during the 1980s. These data were obtained through a series of Right-to-Know requests filed with the respective county boards of elections. Races for down-ballot offices are comprised of races for Attorney General, Auditor General, and Treasurer. Data on votes cast in all national and state elections are available from successive issues of the Pennsylvania Manual. Given that local elections in Pennsylvania are held in odd-numbered years whereas national and state elections are held in even-numbered years, I match each odd-numbered year with the closest prior even-numbered year, for example, matching 1981 with 1980, etc. All regressions include year fixed effects. Regressions in columns (4)–(6) include county fixed effects. Regressions in columns (2) and (5) are estimated on municipalities whose population is smaller than or equal to the median population across all municipalities while regressions in columns (3) and (6) are estimated on municipalities whose population is larger than the median population.

Table A.12: Correlation between Democratic representation on local councils and Democratic vote share in national and state races

	(1)	(2)	(3)	(4)	(5)	(6)
	All municipalities	Smaller municipalities	Larger municipalities	All municipalities	Smaller municipalities	Larger municipalities
Panel A: Share of Democrats on councils regressed on Democratic vote share based on elections for President in 2012						
Democratic vote share in state/ national elections	1.206***	1.264***	1.344***	1.518***	1.377***	1.701***
	(0.149)	(0.125)	(0.184)	(0.0847)	(0.111)	(0.0989)
Number of observations	2625	1314	1311	2625	1314	1311
R^2	0.21	0.20	0.25	0.47	0.41	0.56
Panel B: Share of Democrats on councils regressed on Democratic vote share based on elections to down-ballot offices in 2012						
Democratic vote share in state/ national elections	1.615***	1.601***	1.795***	1.709***	1.574***	1.884***
	(0.114)	(0.0936)	(0.136)	(0.0788)	(0.102)	(0.0932)
Number of observations	2625	1314	1311	2625	1314	1311
R^2	0.36	0.34	0.41	0.52	0.46	0.60
Panel C: Share of Democrats on councils regressed on Democratic vote share based on elections for U.S. Senate in 2010 or 2012						
Democratic vote share in state/ national elections	1.415***	1.453***	1.555***	1.526***	1.370***	1.724***
	(0.133)	(0.0986)	(0.162)	(0.0822)	(0.0991)	(0.0916)
Number of observations	5251	2628	2623	5251	2628	2623
R^2	0.30	0.30	0.33	0.48	0.42	0.56
Panel D: Share of Democrats on councils regressed on Democratic vote share based on elections for Governor in 2010						
Democratic vote share in state/ national elections	1.517***	1.566***	1.632***	1.689***	1.469***	1.963***
	(0.148)	(0.101)	(0.188)	(0.0780)	(0.0947)	(0.0943)
Number of observations	2626	1314	1312	2626	1314	1312
R^2	0.31	0.30	0.34	0.49	0.42	0.59
County FE				✓	✓	✓

Robust standard errors clustered at the county level are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The dependent variable in these regressions is the share of Democrats on the governing bodies of their respective municipalities: mayors and councils in the case of cities and boroughs, boards of commissioners in the case of townships of the first class, and boards of supervisors in the case of townships of the second class. It is defined as the number of Democrats divided by sum of the number of Democrats and the number of Republicans and is constructed based on data from the Pennsylvania Department of Community and Economic Development for 2012. Data on votes cast in all national and state elections are available from successive issues of the Pennsylvania Manual. Races for down-ballot offices are comprised of races for Attorney General, Auditor General, and Treasurer. Regressions in columns (4)–(6) include county fixed effects. Regressions in columns (2) and (5) are estimated on municipalities whose population is smaller than or equal to the median population across all municipalities while regressions in columns (3) and (6) are estimated on municipalities whose population is larger than the median population.